

7 The Application Layer

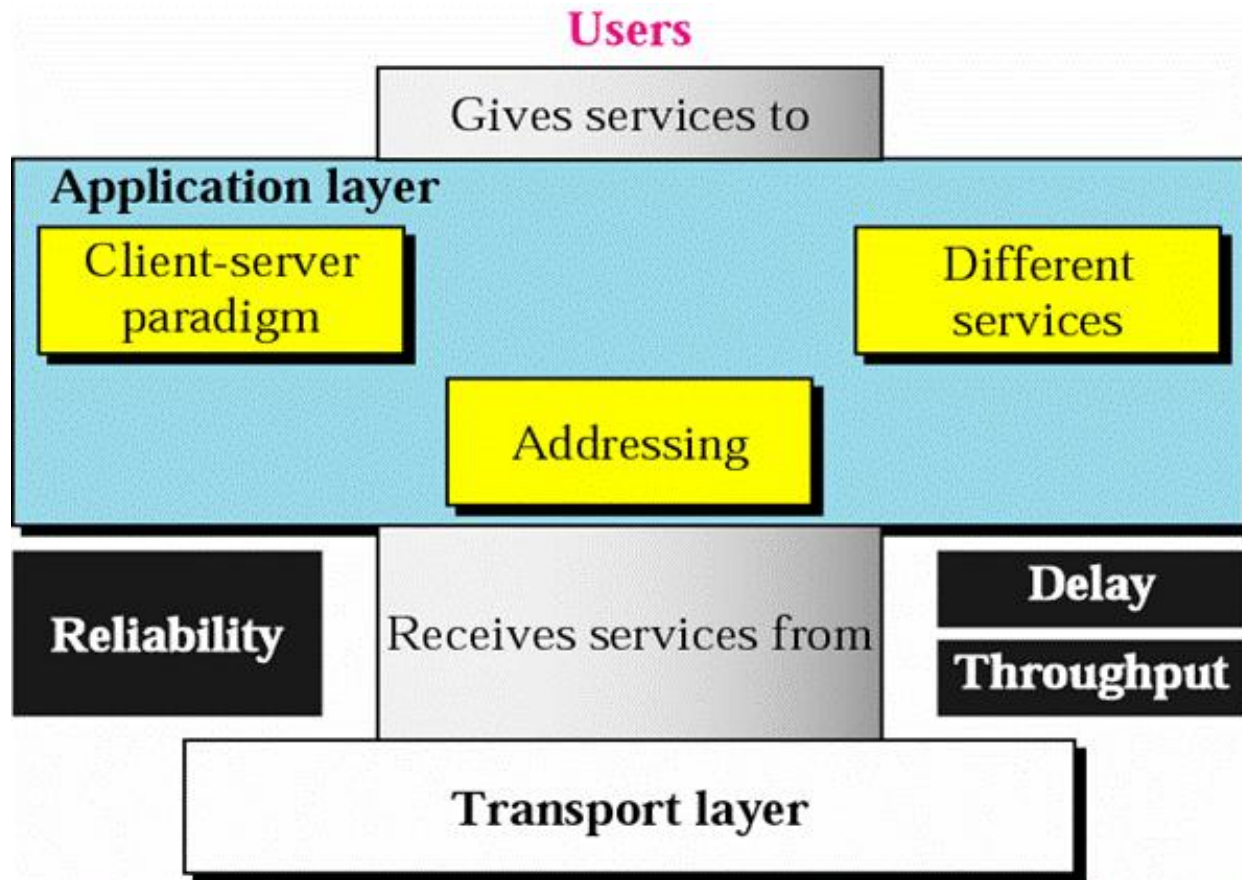
(about 4 hours)

Outline

- Introduction to Application Layer
 - ◆ Positions and functions of application layer
 - ◆ Peer-to-peer paradigm vs. Client-server paradigm
- Domain Name System (DNS)
- Email System and Protocols
- World Wide Web (WWW)

Position of Application Layer

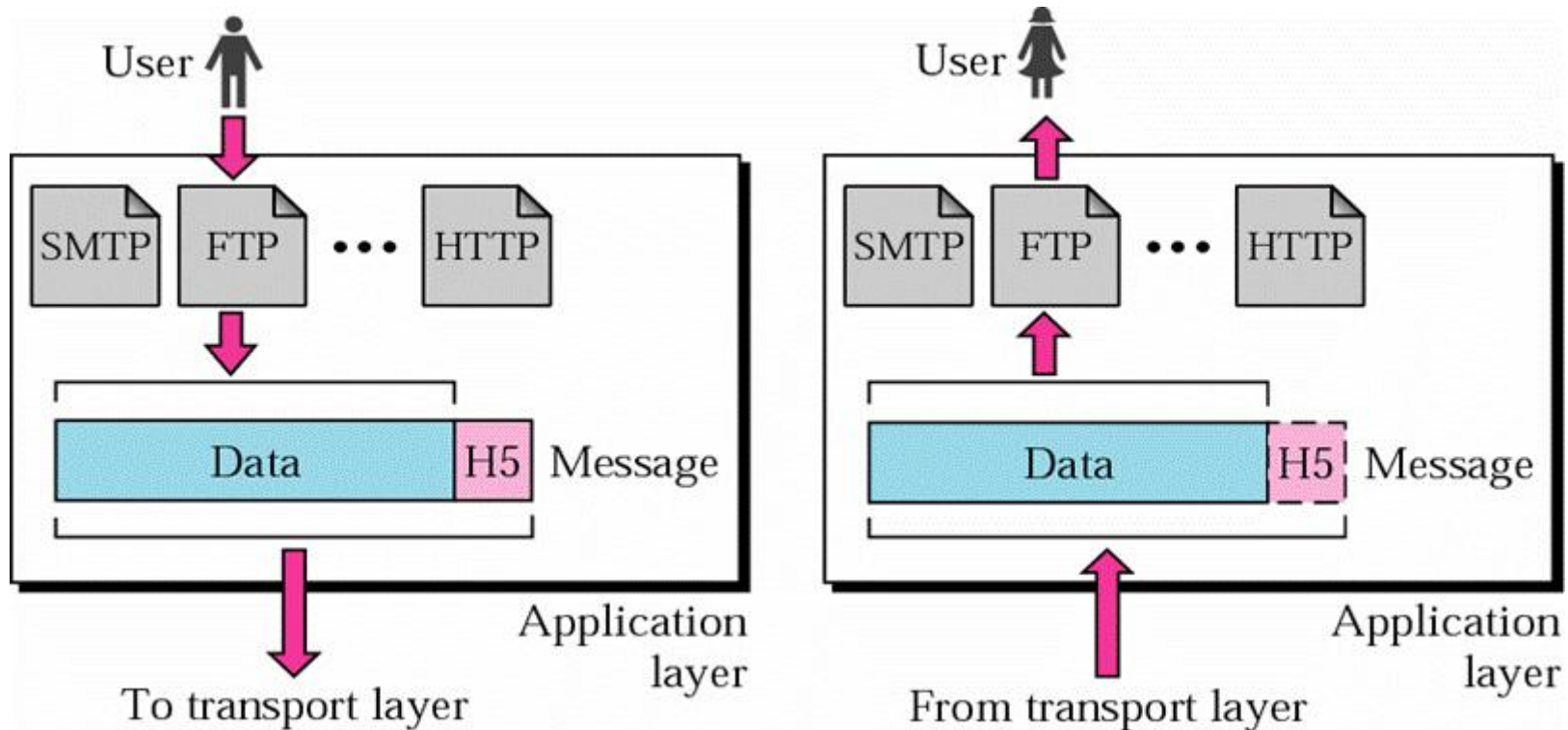
■ Top layer of the Layering Architecture



[Forouzan]

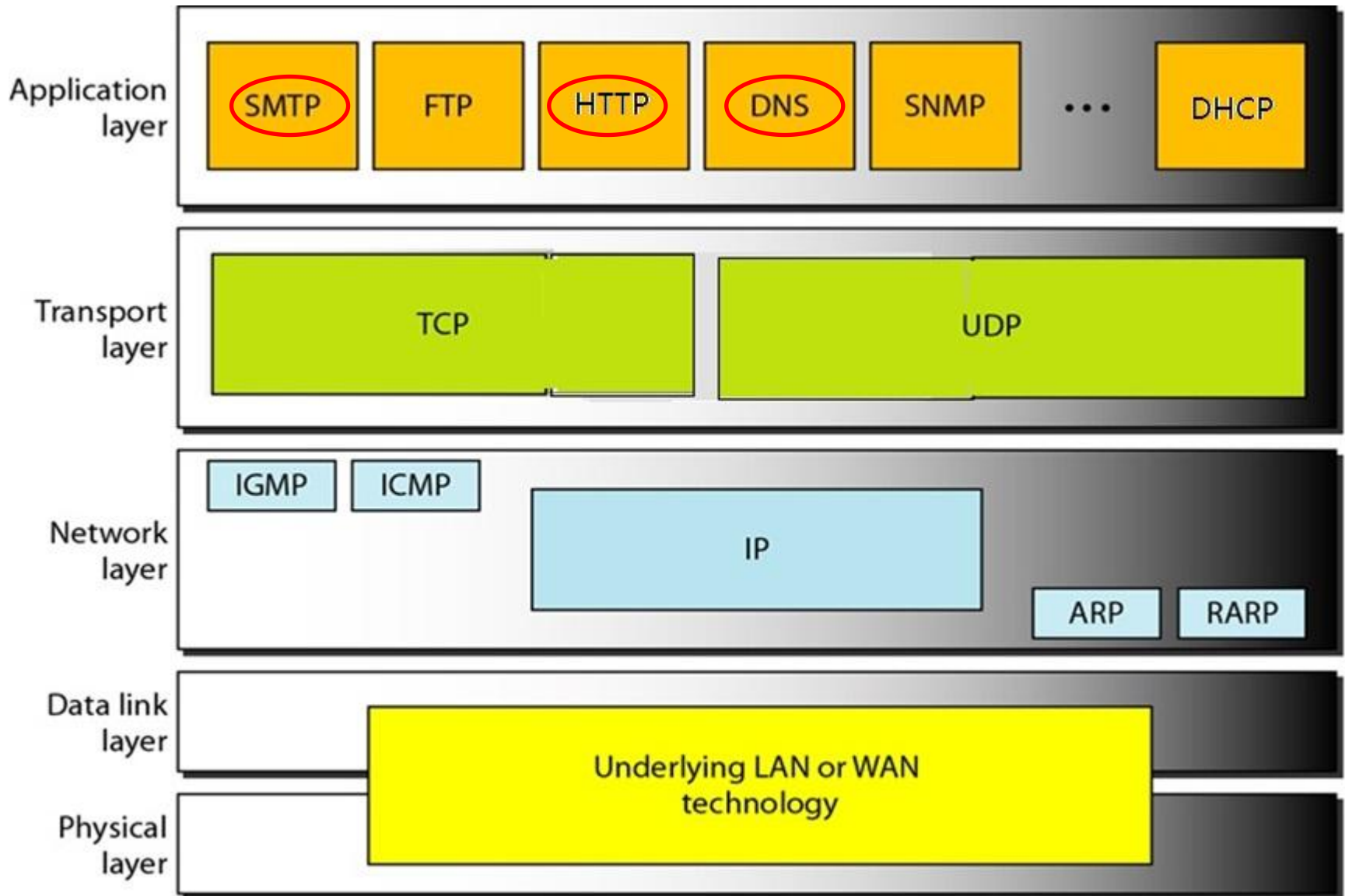
Duties and Features of Application Layer

- Providing various applications to end users
- Maximum number of protocols and most complex
- Each protocol is specified to a type of application, no general-purpose protocols



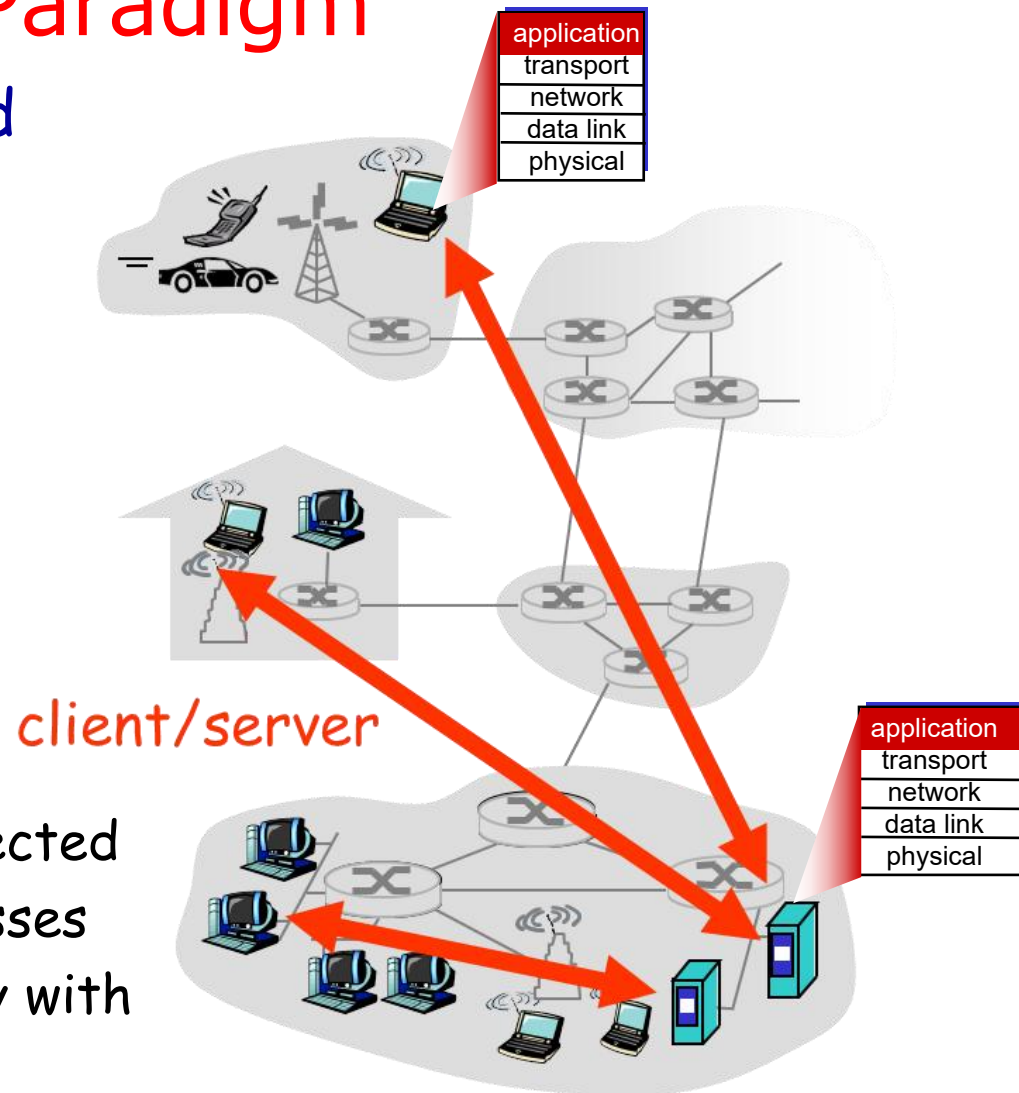
[Forouzan]

Conventional Internet Application Protocols



Client-server Paradigm

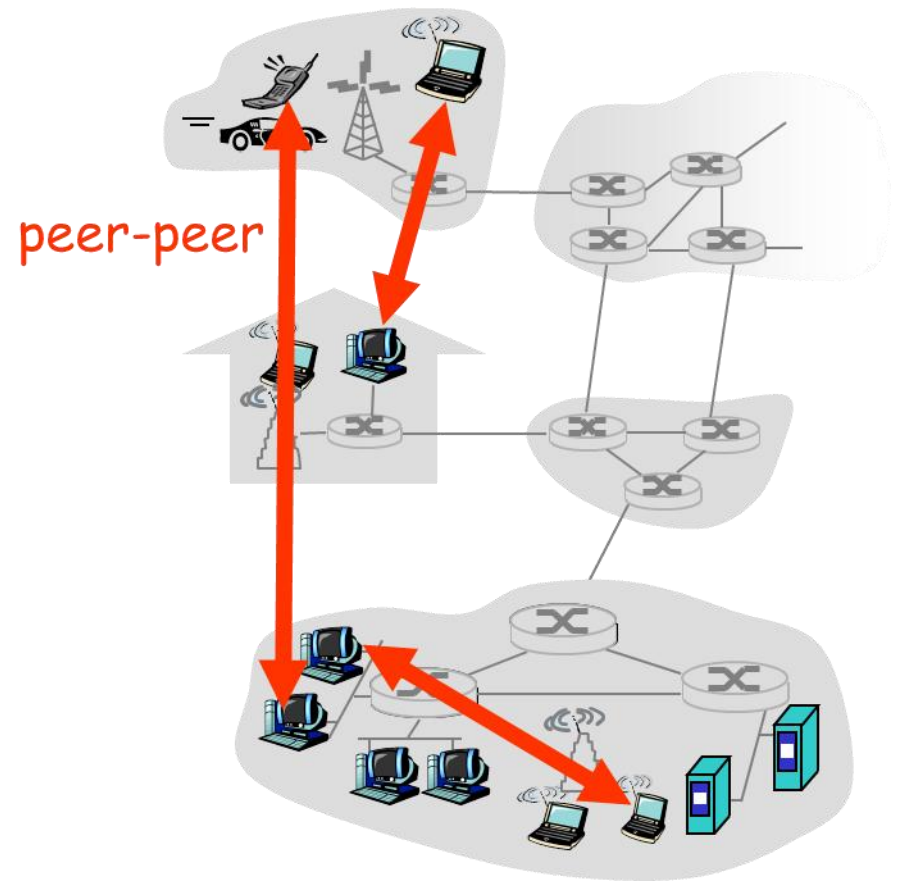
- The tasks, capabilities and protocol machines are different at each side.
- server:
 - ◆ always-on
 - ◆ permanent IP address
 - ◆ server farms for scaling
- clients:
 - ◆ communicate with server
 - ◆ may be intermittently connected
 - ◆ may have dynamic IP addresses
 - ◆ do not communicate directly with each other
- DNS, Web, email...



Forms the basis of computer networks

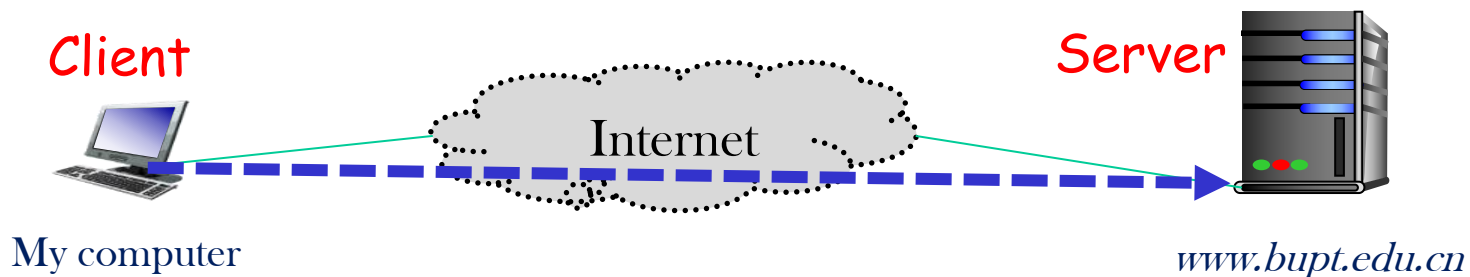
Pure P2P Architecture in Computer Networks

- arbitrary end systems directly communicate
- peers are intermittently connected and change IP addresses
- BT, eMule, 迅雷...



[Kurose]

Application Scenario: 访问北邮主页



■ WWW Application

- ◆ QoS: reliable transmission
- ◆ Transport layer: TCP
- ◆ Step 1: handshake--clients to web service
- ◆ **www.bupt.edu.cn ---> IP address?**

➡ DNS

Outline

- Introduction to Application Layer
- Domain Name System (DNS) (Textbook 7.1)
 - ◆ Hierarchical namespace
 - ◆ Resource records
 - ◆ Name servers
 - ◆ Name Resolution Procedure
- Email System and Protocols
- World Wide Web (WWW)

DNS: Domain Name System(域名系统)

- IP address and Host name

- Name Space

 - ◆ *Flat(平面) Name Space*, each host in the network is unique and independent from each other, Out-dated "**hosts.txt**":

 - *Hierarchical(层次) Name Space*: Host have a composite names that are all hierarchically organized

- DNS

 - ◆ primarily used for mapping **host names(domain name)** and email destinations to IP addresses - but can be used for other purposes

 - ◆ **domain name**: identifies each host on the Internet with a unique user-friendly **name**, e.g. **www.bupt.edu.cn** instead of 123.127.134.10

- Providing support for other applications

 - ◆ Translating **domain name (host name)** to **IP address**

DNS: Domain Name System(域名系统)

■ Host ID in Internet

◆ Host name: **domain name**, eg. www.bupt.edu.cn (**Application**)

DNS

◆ Logical address: **IP address** (**Network-Global**)

ARP

◆ Physical address: **MAC address** (**Link-Local**)

■ DNS Features

◆ **Client/Server** application, DNS Server (port number 53)

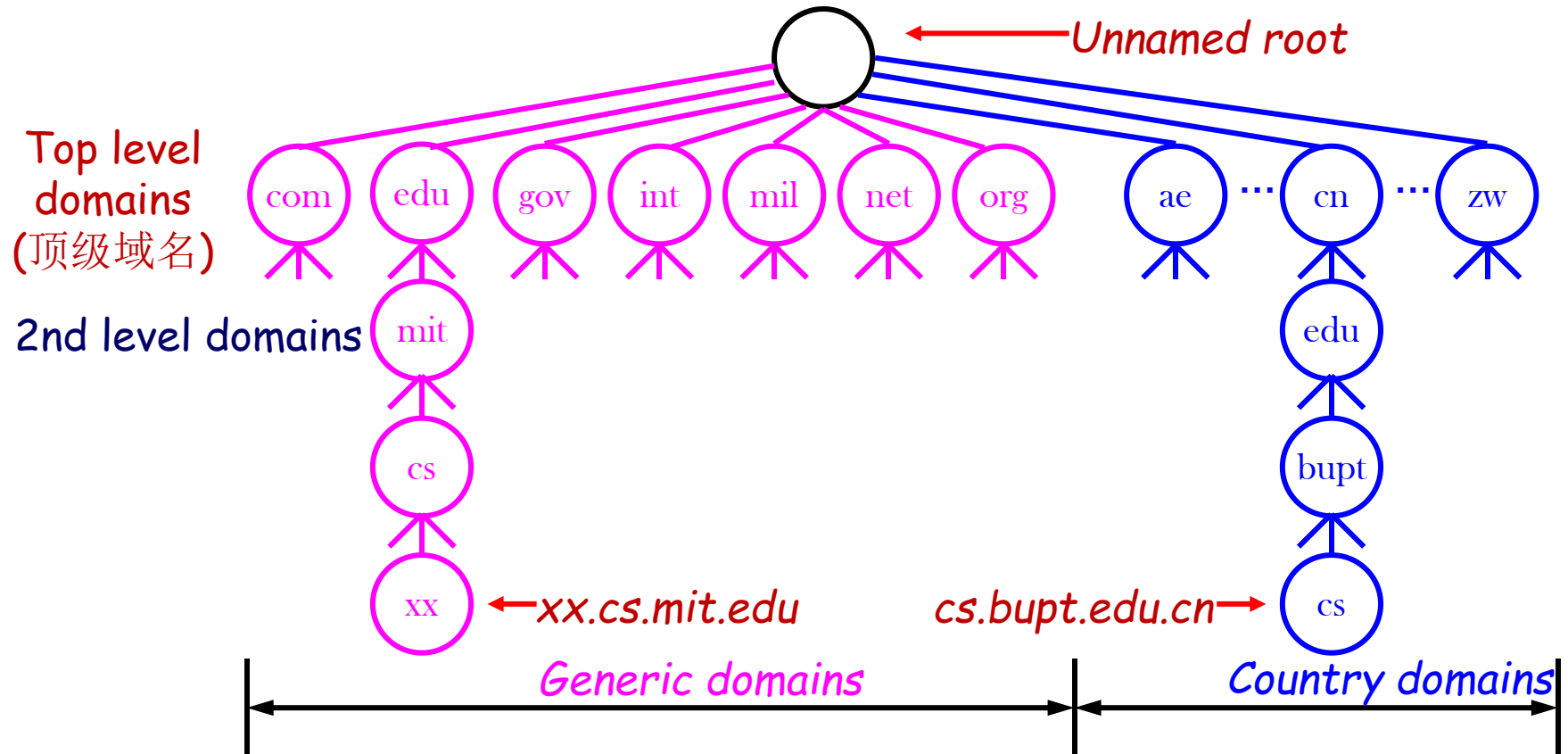
◆ UDP datagram

◆ **Domain** name space--**hierarchical name space**

◆ **Distributed database**

◆ Naming follows organizational boundaries, not physical networks

Hierarchical Name Space



- Within an organization,
 - ◆ Subdivision possible
 - ◆ Arbitrary levels possible (maximum depth = 128)
 - ◆ Not standardized, controlled locally by the organization
- Naming policy: by the path upward from leaf to root, separated by "."

Resource Records (RR)

- Each domain in the DNS can have a set of **resource records**(RRs) associated with it. These records are the DNS database.
- The primary function of DNS is to **map domain names onto resource records**.
- A RR is a five-tuple, the format as follows:

Domain_name Time_to_live Class Type Value

- ◆ Domain_name: tells the domain to which this record applies
- ◆ TTL: specifies the Time To Live (in unit of second) of the **cached RRs**
- ◆ Type: tells what kind of record this is (Figure7-3)
 - A – IPv4 address of a host(32-Bit)
 - AAAA – IPv6 address of a host(128-Bit)
 - NS – Name Server
 - CNAME – Canonical Name, alias
- ◆ Class: specifies the protocol family to use
 - IN – the Internet information
- ◆ Value: depend on type

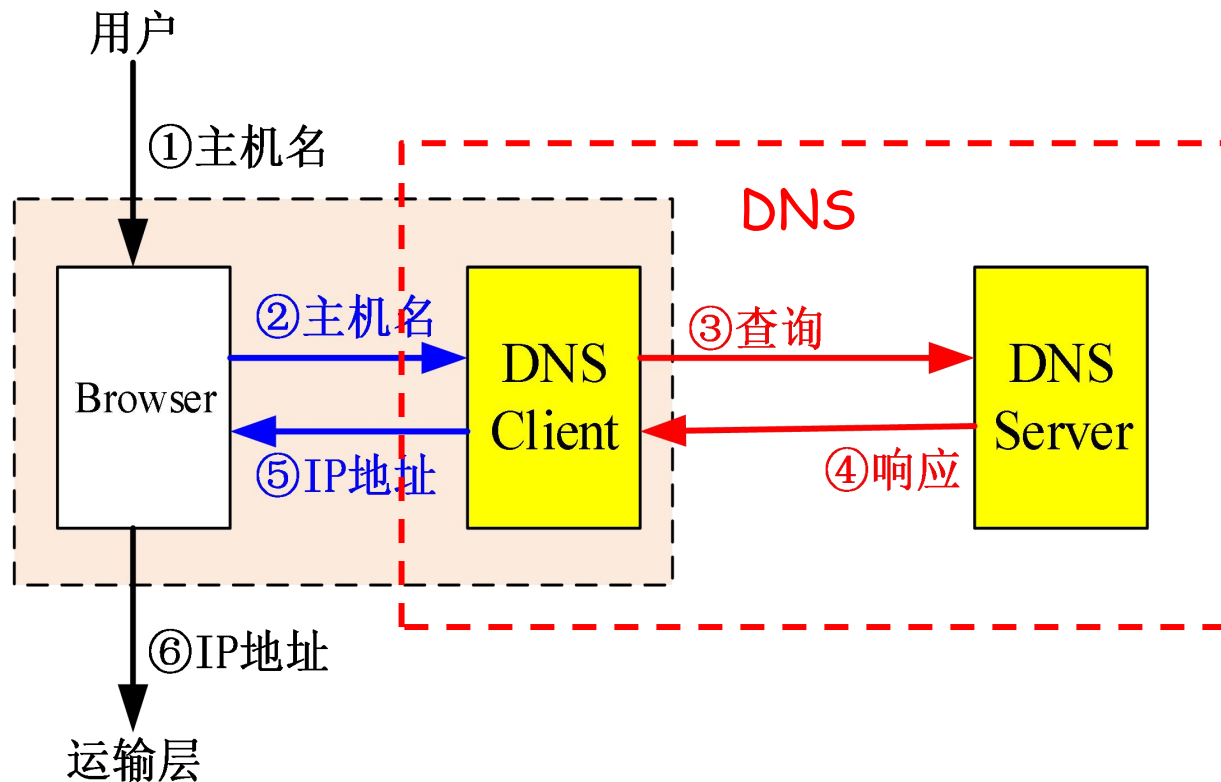
Sample of DNS Database

Domain name	Time-to-live	Class	Type	Value
; Authoritative data for cs.vu.nl				
cs.vu.nl.	86400	IN	SOA	star boss (9527,7200,7200,241920,86400)
cs.vu.nl.	86400	IN	MX	1 zephyr
cs.vu.nl.	86400	IN	MX	2 top
cs.vu.nl.	86400	IN	NS	star
star	86400	IN	A	130.37.56.205
zephyr	86400	IN	A	130.37.20.10
top	86400	IN	A	130.37.20.11
www	86400	IN	CNAME	star.cs.vu.nl
ftp	86400	IN	CNAME	zephyr.cs.vu.nl
flits	86400	IN	A	130.37.16.112
flits	86400	IN	A	192.31.231.165
flits	86400	IN	MX	1 flits
flits	86400	IN	MX	2 zephyr
flits	86400	IN	MX	3 top
rowboat		IN	A	130.37.56.201
		IN	MX	1 rowboat
		IN	MX	2 zephyr
little-sister		IN	A	130.37.62.23
laserjet		IN	A	192.31.231.216

DNS function

■ DNS function

- ◆ Host name map to IP address



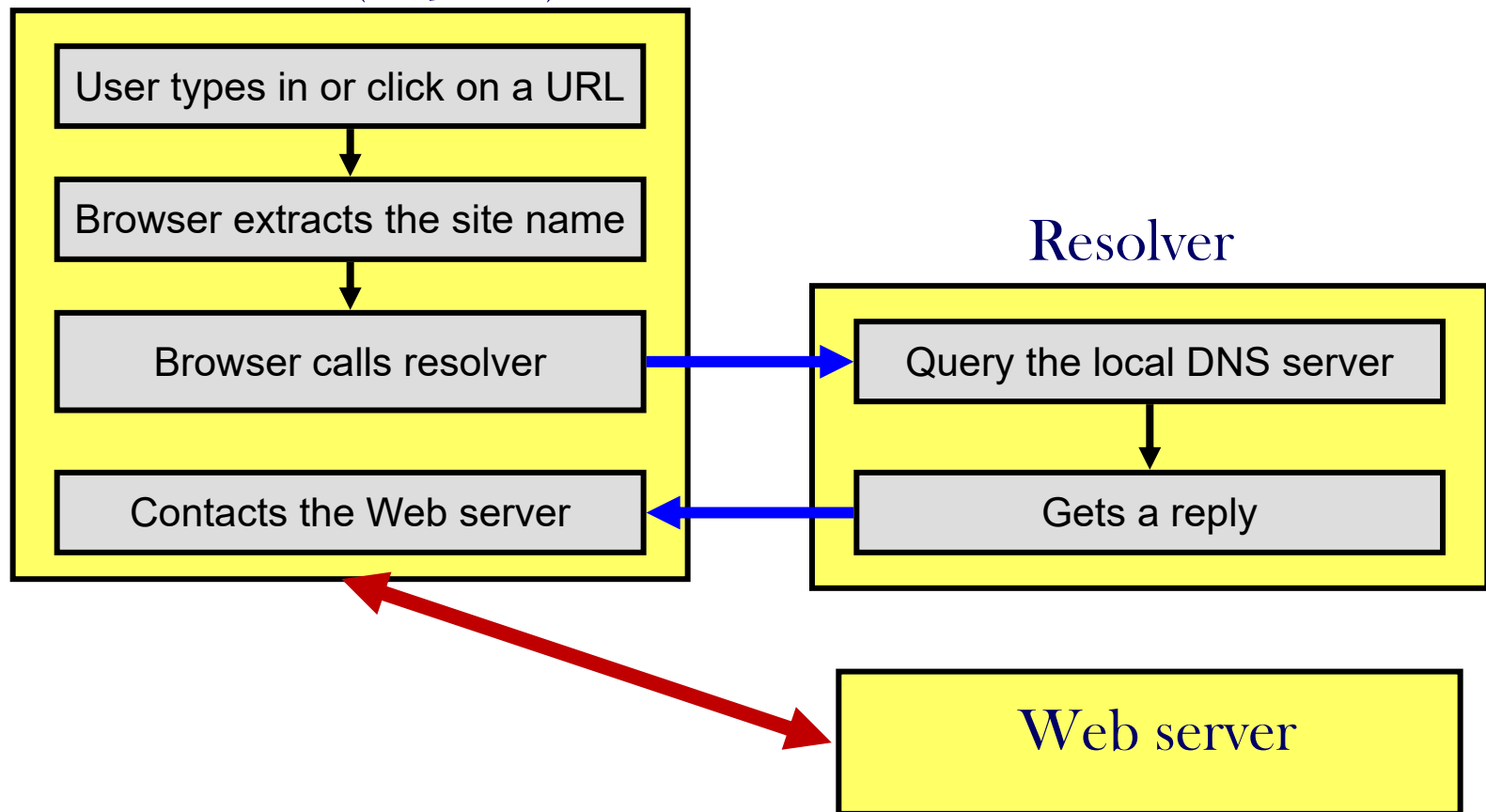
■ DNS Client and DNS Server

DNS Client

■ Client: Known as **resolver**(解析器)

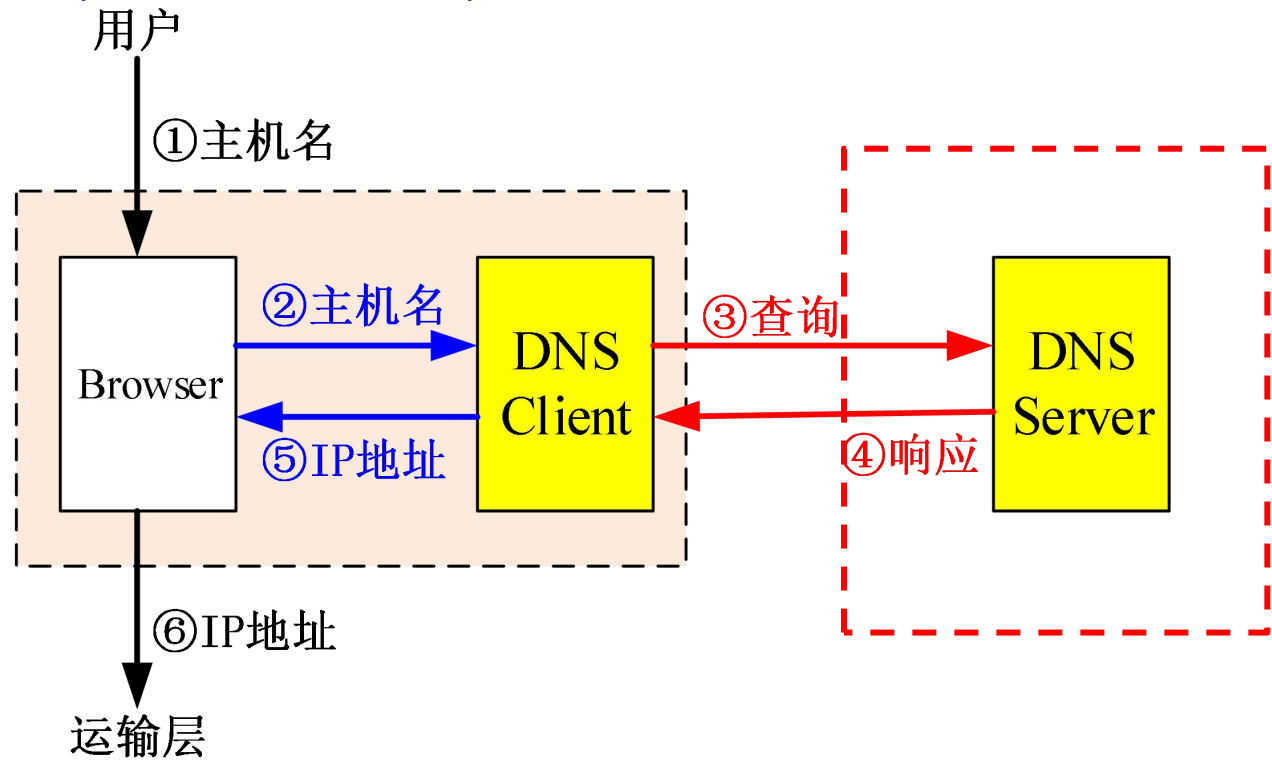
- ◆ software running on client, able to access at least one name server, i.e. **Local name server**(本地名字服务器)

Browser(浏览器)



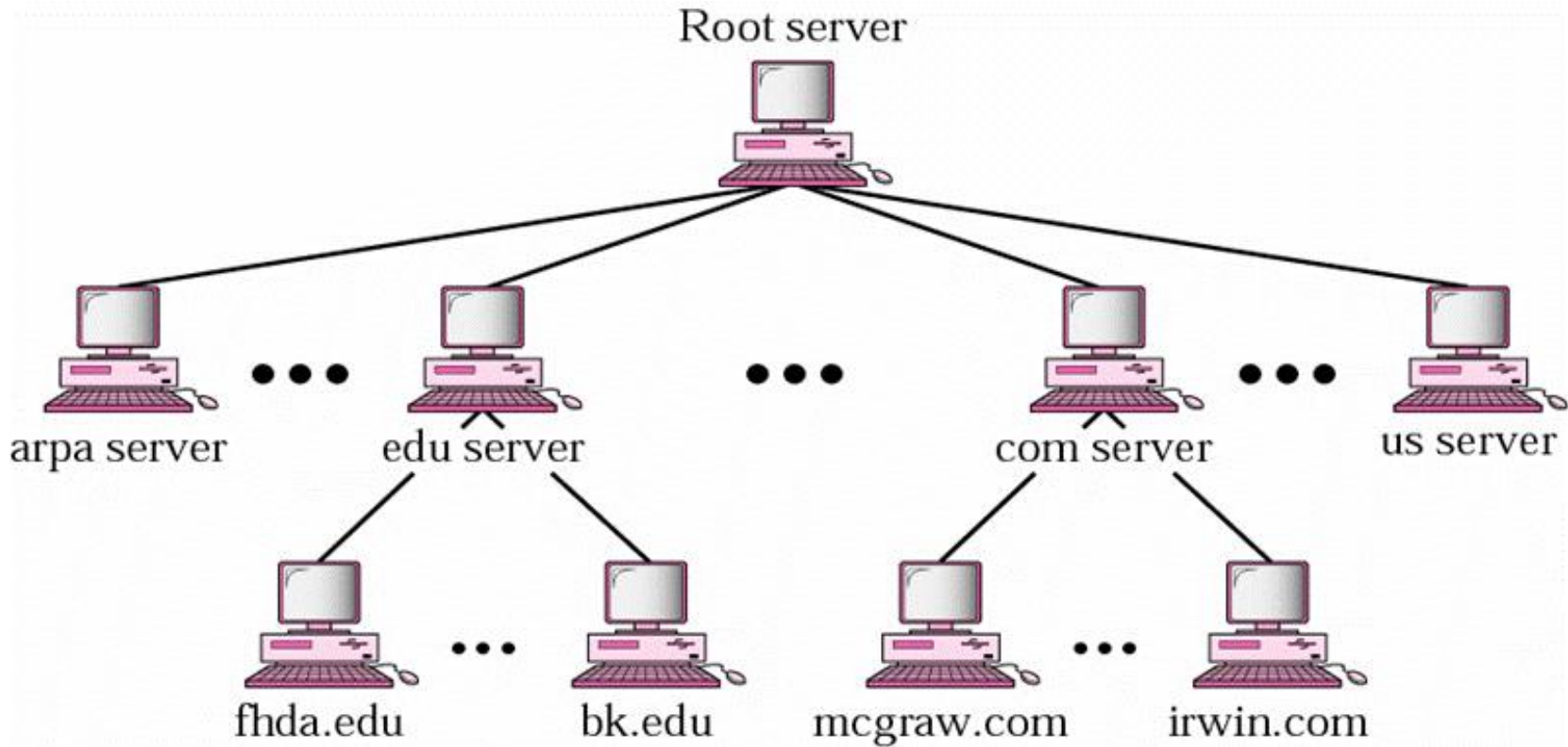
DNS Servers

■ Hierarchical(层次化组织)



- ◆ a single point of failure
- ◆ traffic volume
- ◆ distant centralized database
- ◆ maintenance

Hierarchy of Name Servers

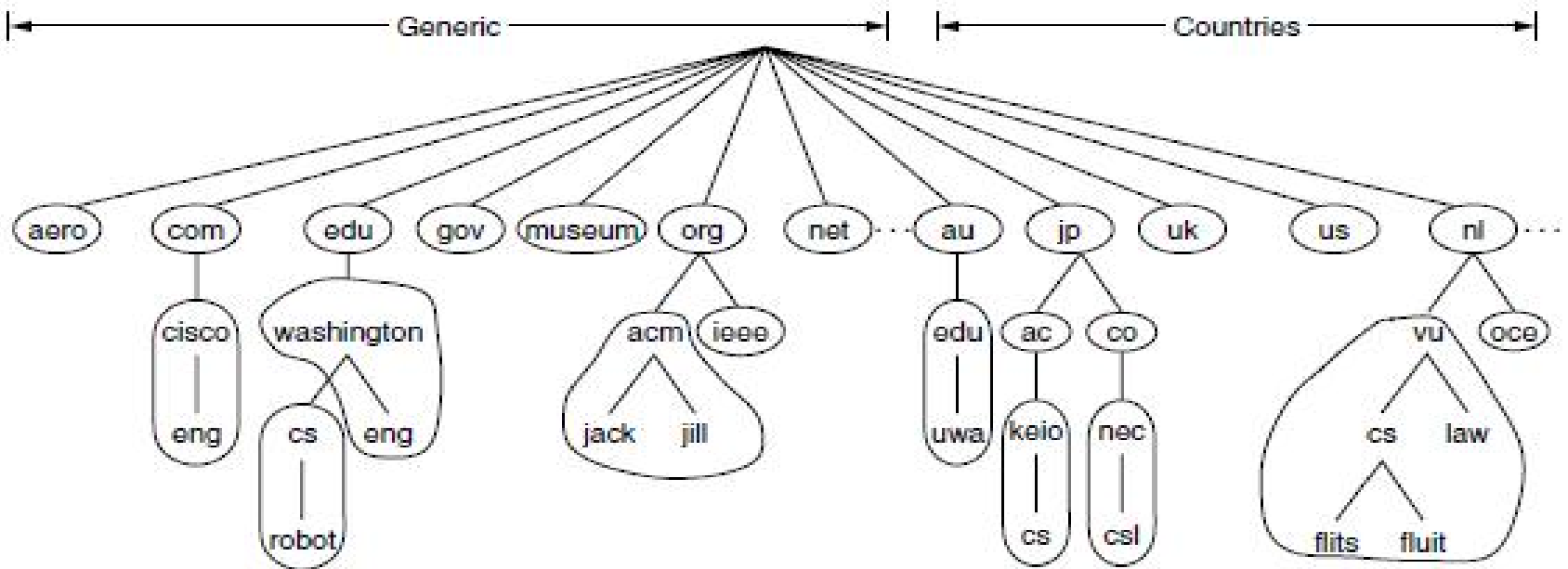


DNS servers are arranged in hierarchy

[Forouzan]

DNS Servers

- DNS name space is divided into nonoverlapping **zones**.
- Each circled **zone** contains some of the tree
- A **zone** corresponds to an administrative authority that is responsible for that portion of the hierarchy e.g. **bupt** controls **.bupt.edu.cn*
- Each zone is associated with one or more name servers. These are hosts that **hold the database for the zone**.



DNS Servers

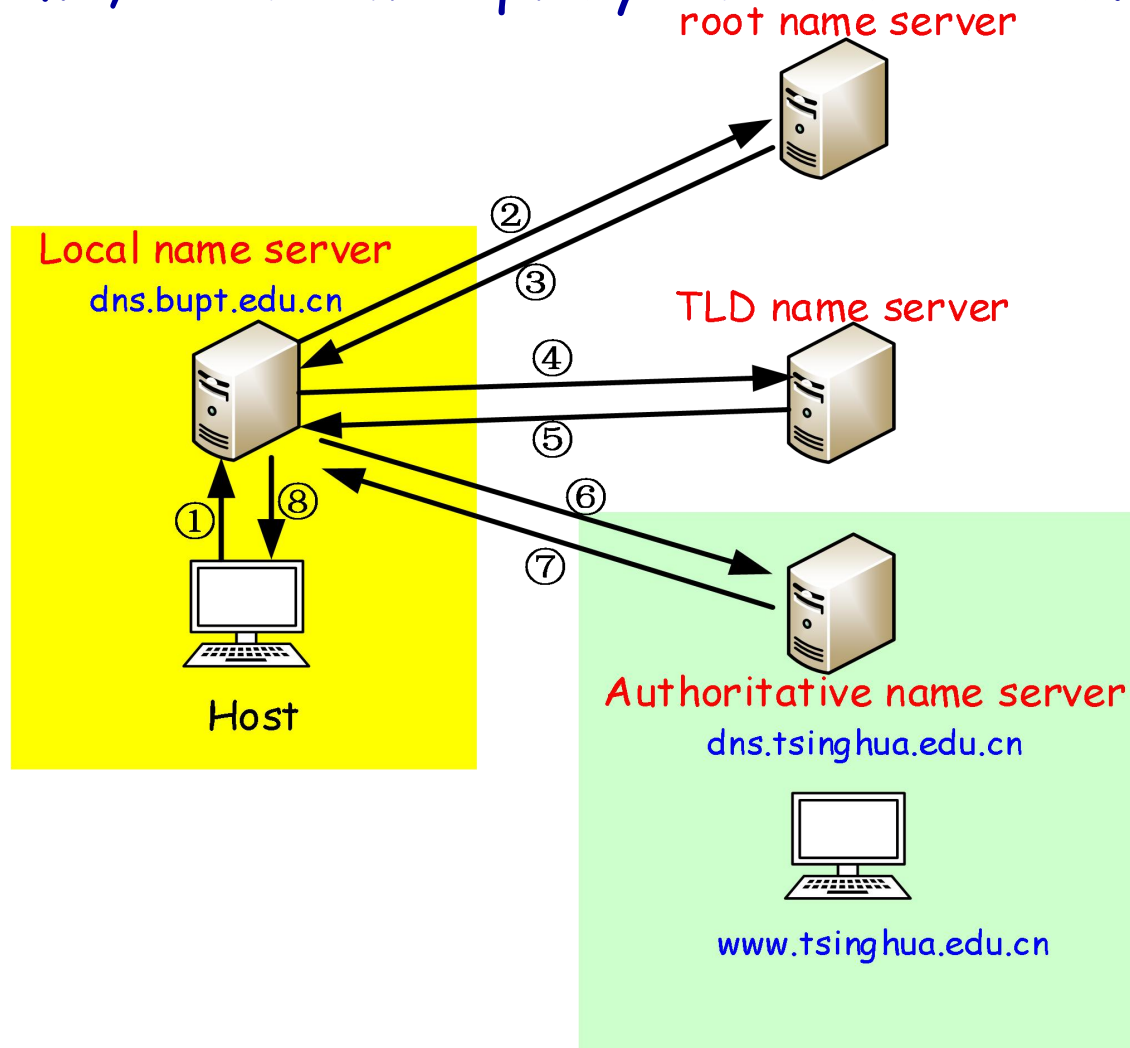
- **Authoritative Name Server**(权威域名服务器)
 - ◆ contains all the records for the hosts in its **zone**
 - ◆ gets records information from a file on its disk
 - ◆ **Authoritative record** is one that comes from the authority that manages the record and is thus **always correct**.
- **How does a server know about other servers that are responsible for the other zones?**
 - ◆ Every server knows the root
 - ◆ Root server knows about all TLD servers
 - ◆ Every server knows the server(s) further down the hierarchy
- **local name Server** (本地域名服务器/默认域名服务器)

Name Resolution

- The process of looking up a name and finding an address is called **name resolution**.
- Applications
 - ◆ Becoming DNS client
 - ◆ Sending request first to local name server
- Local name server
 - ◆ Listening at port 53, normally using UDP
 - ◆ If answer known, returns response
 - ◆ If answer unknown
 - Requests root or TLD name server
 - Follows links
 - Until reaching **Authoritative Name Server**, which performs name translation and returns response

Name Resolution

- A **resolver** passes the query about a domain name to a **local name server**. If local name server can not resolve the name, then it will query another server.



Name Resolution Methods

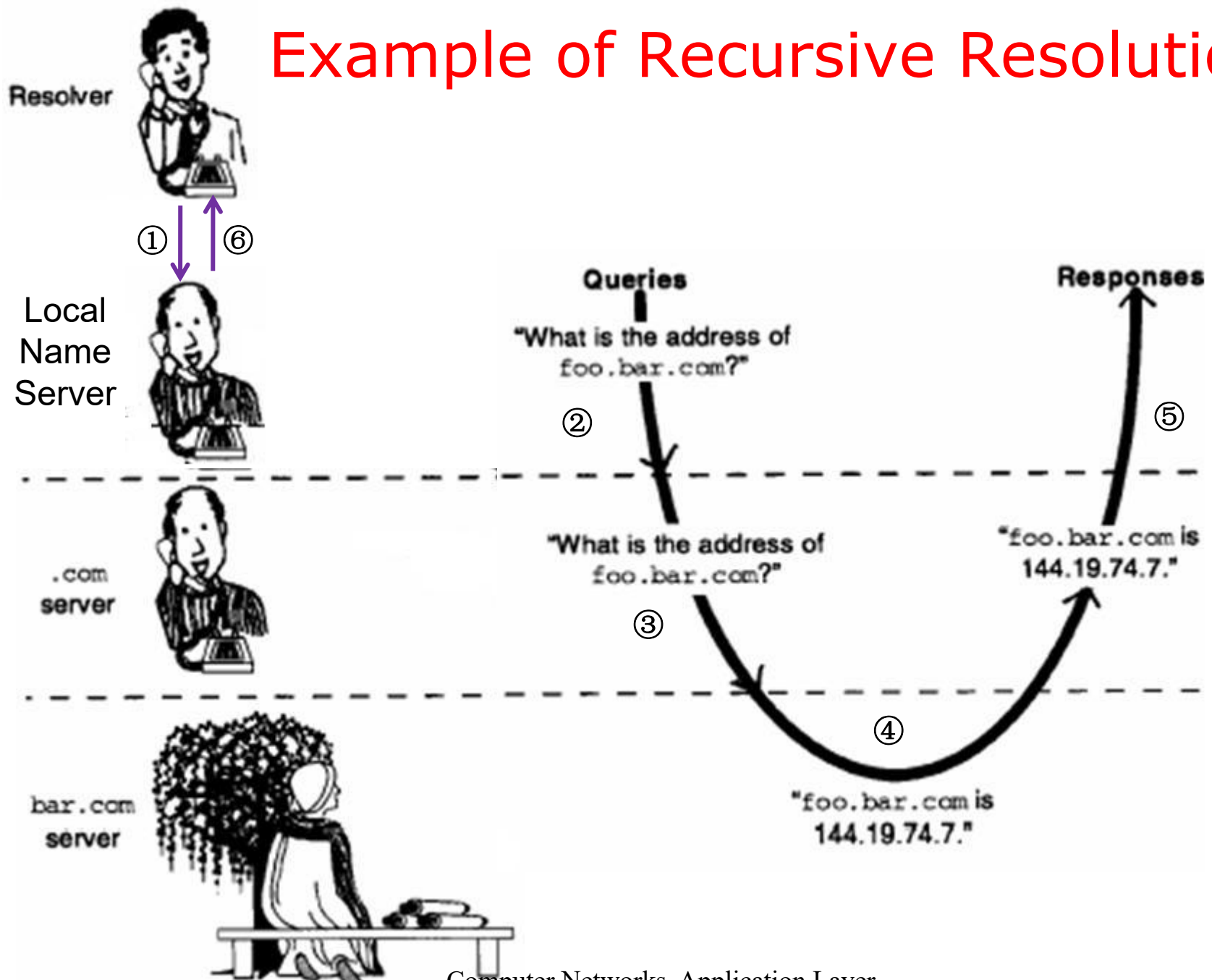
■ Recursive resolution(递归解析/递归查询):

- ◆ If the **queried server** does not have the information, it must **make** the appropriate **query** or queries to get the information
- ◆ The **queried server** performs another recursive query until it has the **desired answer** to return. It **does not return partial answers**.

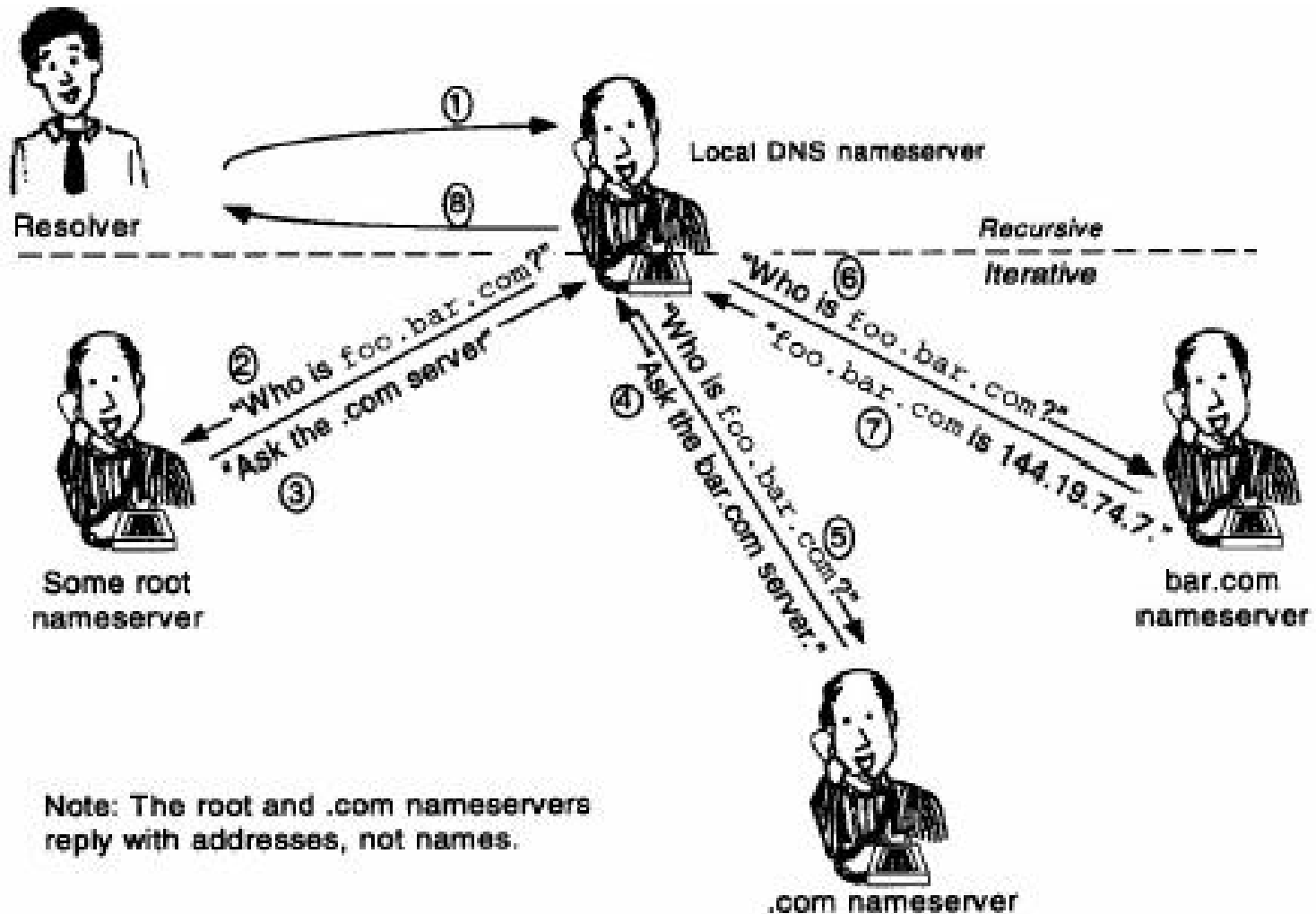
■ Iterative resolution(迭代解析/迭代查询):

- ◆ If the queried server does not have the information, it may then respond with the **address** of another server (**partial answers**); the **local name server** (on behalf of resolver)then **queries** that server (which might respond with the address of another server, and so on)
- ◆ Commonly used by name servers on the Internet

Example of Recursive Resolution

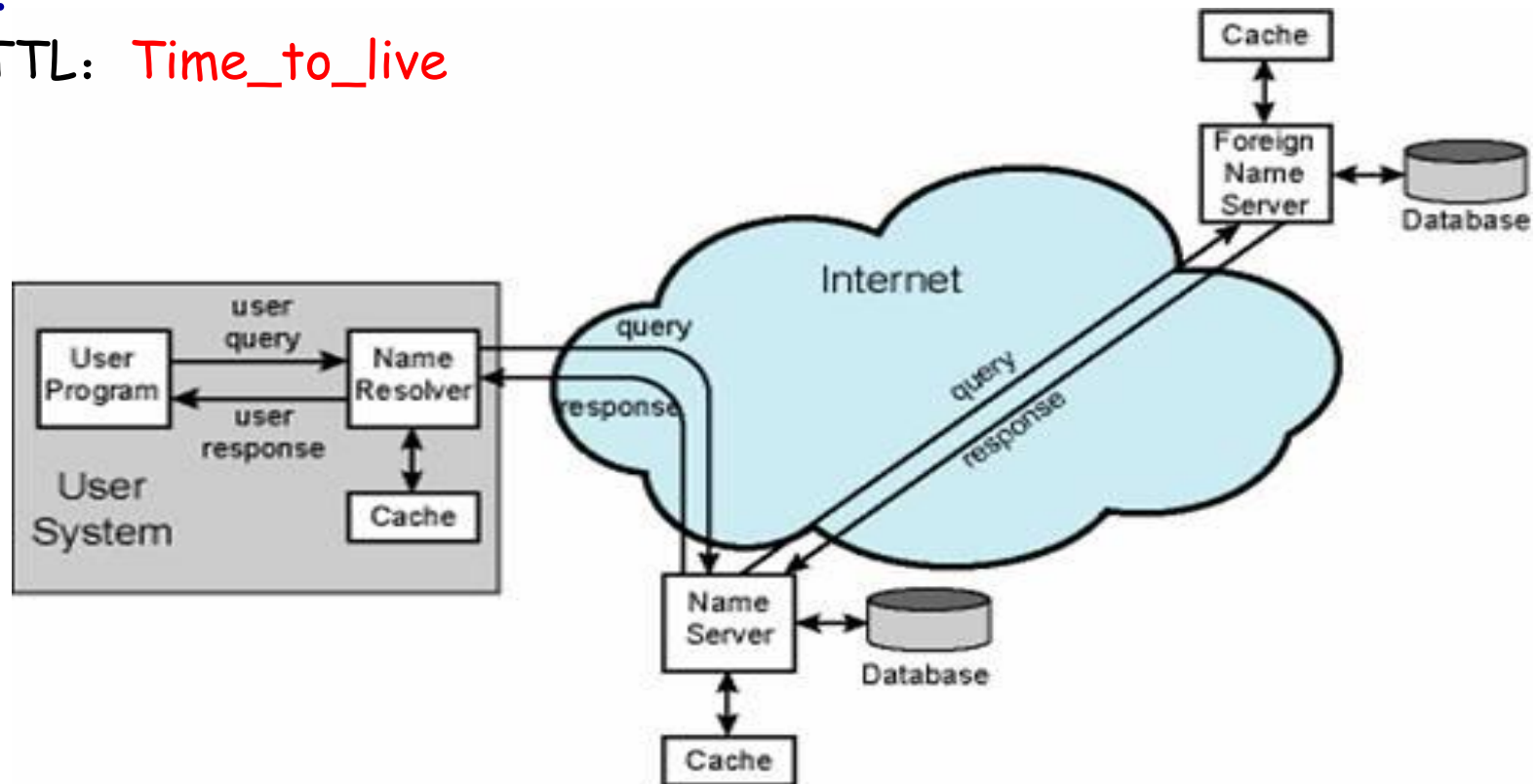


Example of Iterative Resolution



DNS Caching

- All of the **answers**, including all the **partial answers** returned.
 - ◆ Server always caches answers
 - ◆ Host can cache answers
- Using cached answers greatly reduces the steps in a query and improves performance
- Cached answers are **not authoritative**, cache entries should not live too long.
 - ◆ TTL: **Time_to_live**



DNS

- 目标/功能：实现主机域名和IP地址(RR)之间的映射
- 域名
 - ◆ 分层的域名空间（易扩展）、倒置的树
 - ◆ 根域、顶级域名、二级...
- 资源记录RR
 - ◆ 布式数据库, 五元组(Domain_name, TTL, Class, Type, Value)
- Client(解析器)请求本地域名服务器（UDP， 端口53）
- 本地名字服务器请求其他服务器(权威域名服务器)
 - ◆ 迭代解析（因特网普遍采用）
 - ◆ 递归解析
- DNS目标很简单，但实现系统庞大而复杂
- 具有高效率(缓存机制)和可靠性(镜像、辅助域名服务器)
- 安全性(略): DNSSEC, DoT (DNS over TLS), DoH (DNS over HTTPS)

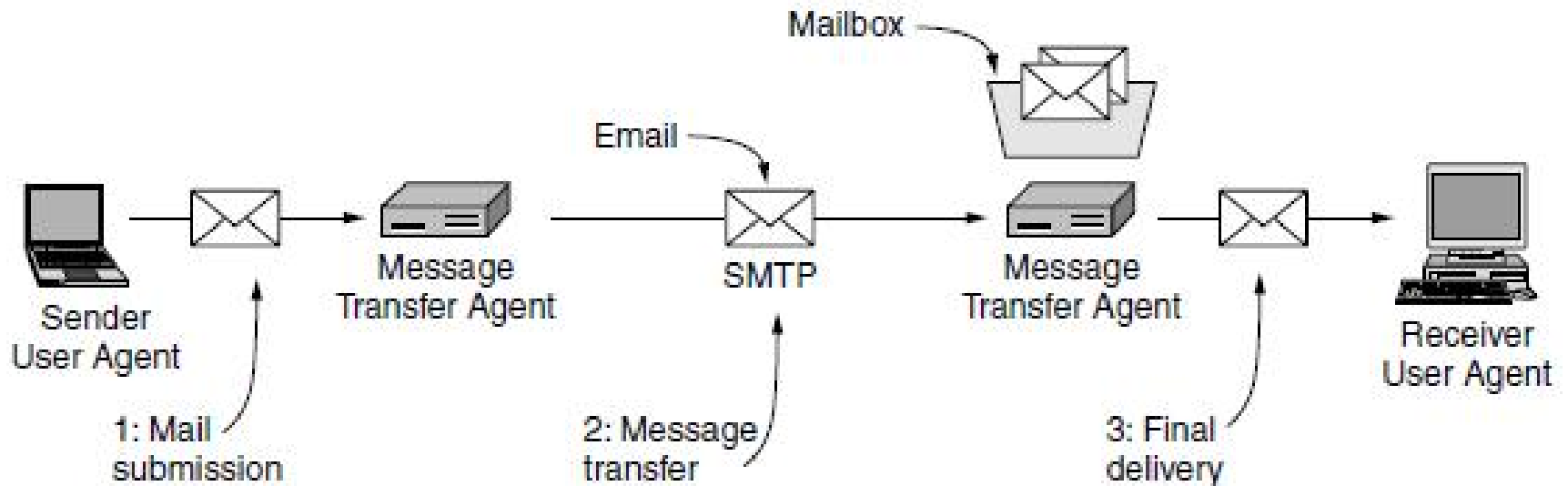
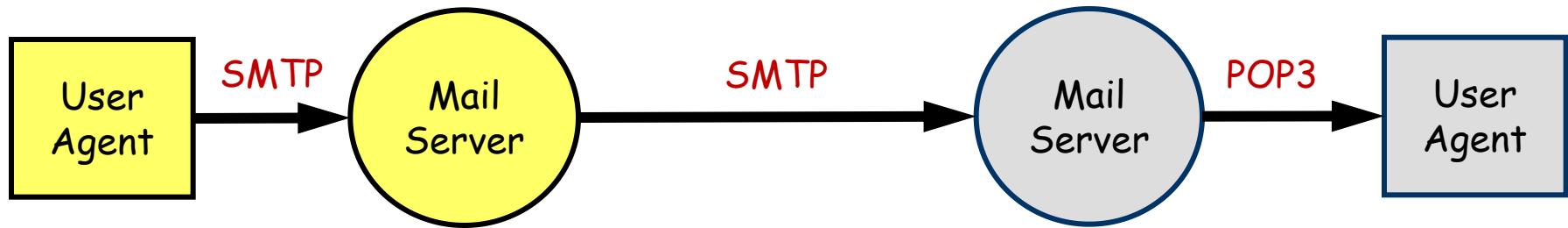
Outline

- Introduction to Application Layer
- Domain Name System (DNS)
- Email System and Protocols
 - ◆ System components
 - ◆ Message Format, MIME
 - ◆ SMTP
 - ◆ POP3, IMAP
- World Wide Web (WWW)

What is Email?

- Electronic Mail (email, e-mail)
- Provides a means to send electronic messages from one person to another **asynchronously**
- One of the most popular applications on the Internet and one of the most important communication methods today
- Email service can be provided by
 - ◆ ISPs: @126.com, @163.com, @sina.com, yahoo.com.cn, ...
 - ◆ Corporations and institutes: @baidu.com, @bupt.edu.cn, @ietf.org, ...
 - ◆ Bundled with other services: @139.com, @qq.com, ...

Components Of Email System (1)



Components Of Email System (2)

■ UA (User Agent)

- ◆ end-user mail program
- ◆ **Interface** between the end users and the email servers
- ◆ E.g., outlook, foxmail, ...

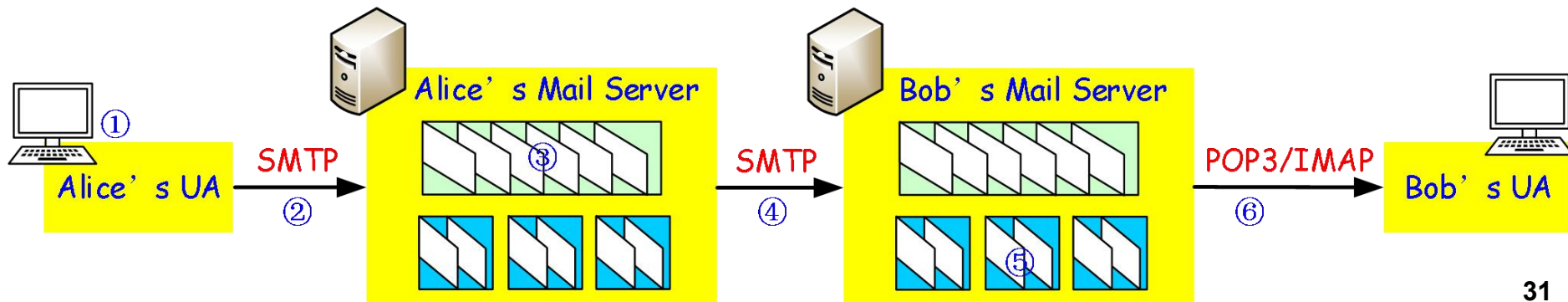
■ MTA(Message Transfer Agent)—Mail Server

- ◆ Responsible for **transmitting/receiving emails** and reporting status information about mail transferring to the mail sender
- ◆ **Both a client and a server**

■ Mailbox: store email

■ Email protocols

- ◆ SMTP: used for sending an email
- ◆ POP3/IMAP: used for receiving an email



Basic Functions Of Email System

- **Composition** - refers to the process of creating messages and answers.
- **Transfer** - refers to moving messages from the originator to the recipient.
- **Reporting** - refers to the process of informing the originator what happened to the message.
- **Displaying** - means of showing the messages.
- **Disposition(处置)** - refers to what happened to the message after it has been read by the receiver.

Email Address

- The destination address must be in a format that the message transfer agents can deal with
- The expected form of an address is *user@dns-address* (RFC 5322)
- Each email address is *unique* on the Internet because
 - ◆ *dns-address* is unique on the Internet
 - ◆ *user* is unique in the dns-address

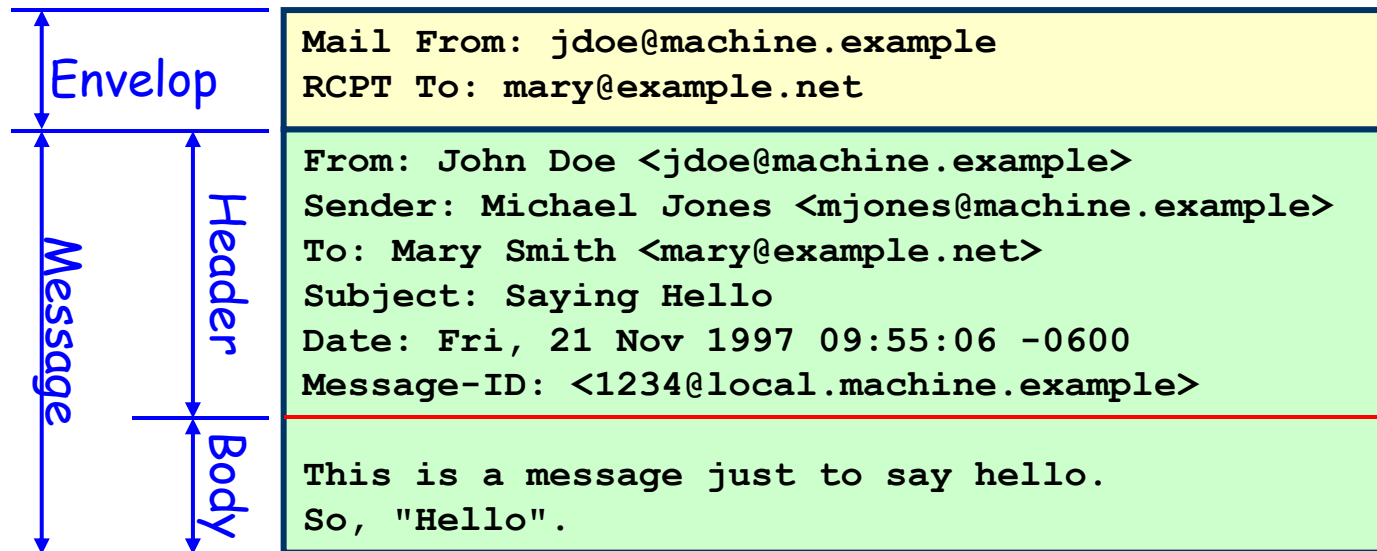
Internet Message Format(RFC 5322)

■ Message envelop

- ◆ Contains whatever information is needed to accomplish transmission and delivery
- ◆ Constructed by **Message Transfer Agent**

■ Message contents

- ◆ comprise the object to be delivered to the recipient
- ◆ Headers: from, to, subject, date, postmarks
- ◆ **Blank line**
- ◆ Body: actual message, may have many parts

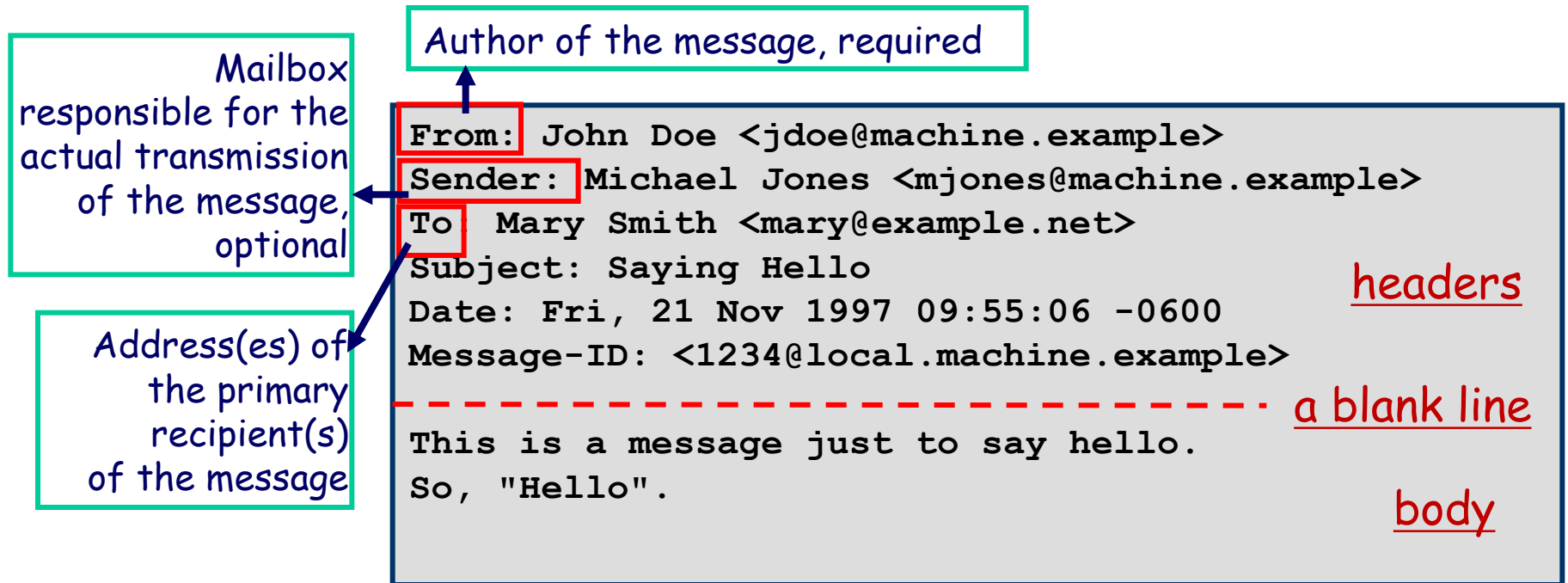


Message Field Definitions

Categories	Header	Meaning
Originator fields	From:	Person or people who created the message
	Sender:	Email address of the actual sender
	Reply-to:	Email address to which replies should be sent
Destination address fields	To:	Email address(es) of primary recipient(s)
	Cc:	Email address(es) of secondary recipient(s)
	Bcc:	Email address(es) for blind carbon copies
The origination date field	Date:	The date and time the message was sent
Identification fields	Message-Id:	Unique number for referencing this message later
	References:	Other relevant Message-Ids
Information fields	Subject:	Short summary of the message for the one-line display
	Keywords:	User chosen keyword
Trace fields	Received:	Line added by each transfer agent along the route
	Return-Path:	Can be used to identify a path back to the sender

Example of Mail Message

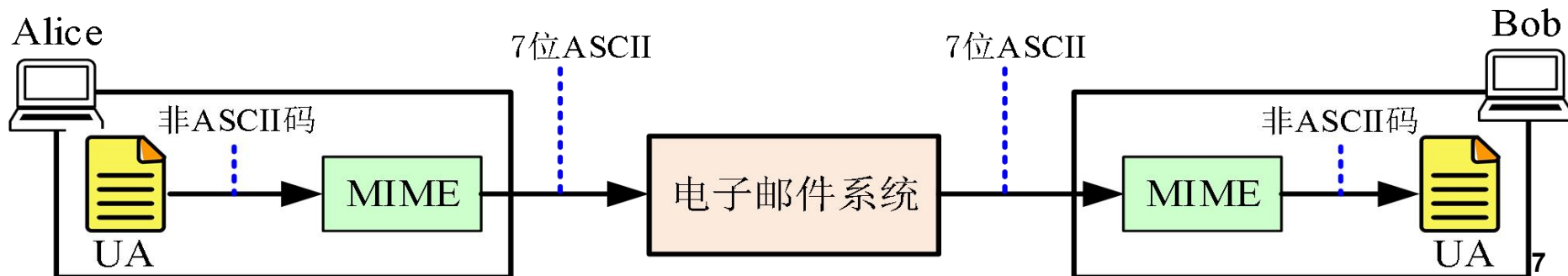
- The **header** is everything up to the blank line, and the **body** is everything after the blank line



- Problem of RFC 822: **Only 7 bit ASCII format** can be used

MIME

- Multipurpose Internet Mail Extension, widely used now
- Extension for multipart & multimedia email
- The basic idea of MIME is to continue to use the RFC822 format but to add structure to the message body and define encoding rules for the transfer of non-ASCII message.
- Supports multipart message content type
 - ◆ each part has its own type and encoding
- MIME messages can be sent using the existing mail programs and protocols
- Widely used now



MIME: New Headers

Header	Meaning
MIME-Version:	Identifies the MIME version
Content-Description:	Human-readable string telling what is in the message
Content-Id:	Unique identifier
Content-Transfer-Encoding:	How the body is wrapped for transmission
Content-Type:	Type and format of the content

MIME: Content Types and Subtypes

Type	Subtype	Description
Text	Plain	Unformatted text
	Enriched	Text including simple formatting commands
Image	Gif	Still picture in GIF format
	Jpeg	Still picture in JPEG format
Audio	Basic	Audible sound
Video	Mpeg	Movie in MPEG format
Application	Octet-stream	An uninterpreted byte sequence
	Postscript	A printable document in PostScript
Message	Rfc822	A MIME RFC 822 message
	Partial	Message has been split for transmission
	External-body	Message itself must be fetched over the net
Multipart	Mixed	Independent parts in the specified order
	Alternative	Same message in different formats
	Parallel	Parts must be viewed simultaneously
	Digest	Each part is a complete RFC 822 message

Example of a Multipart Message(1)

From: alice@cs.washington.edu
To: bob@ee.uwa.edu.au
MIME-Version: 1.0
Message-Id: <0704760941.AA00747@cs.washington.edu>
Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
Subject: Earth orbits sun integral number of times

This is the preamble. The user agent ignores it. Have a nice day.

--qwertyuiopasdfghjklzxcvbnm
Content-Type: text/html

<p>Happy birthday to you

Happy birthday to you

Happy birthday dear Bob

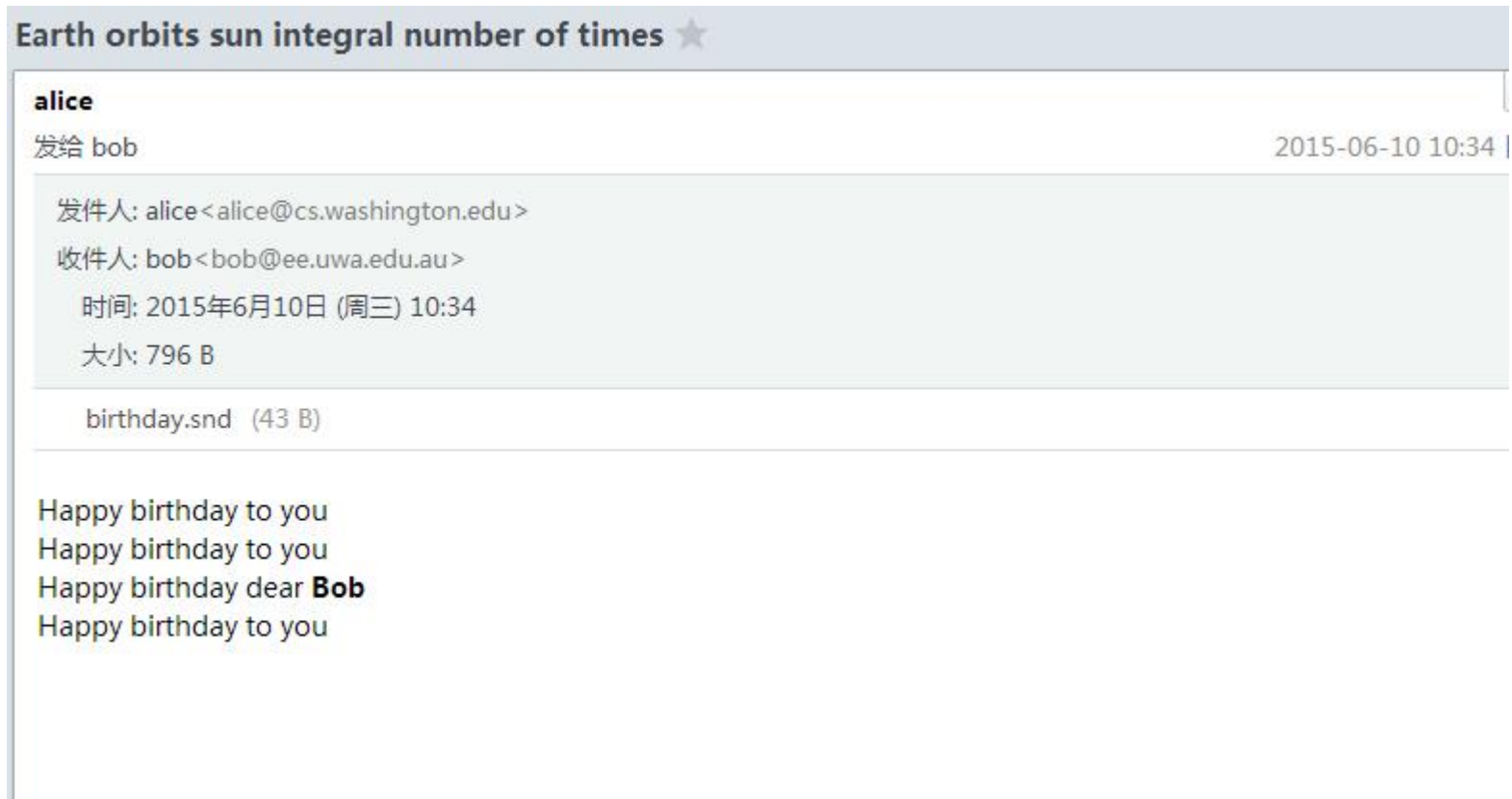
Happy birthday to you</p>

--qwertyuiopasdfghjklzxcvbnm
Content-Type: message/external-body;
 access-type="anon-ftp";
 site="bicycle.cs.washington.edu";
 directory="pub";
 name="birthday.snd"

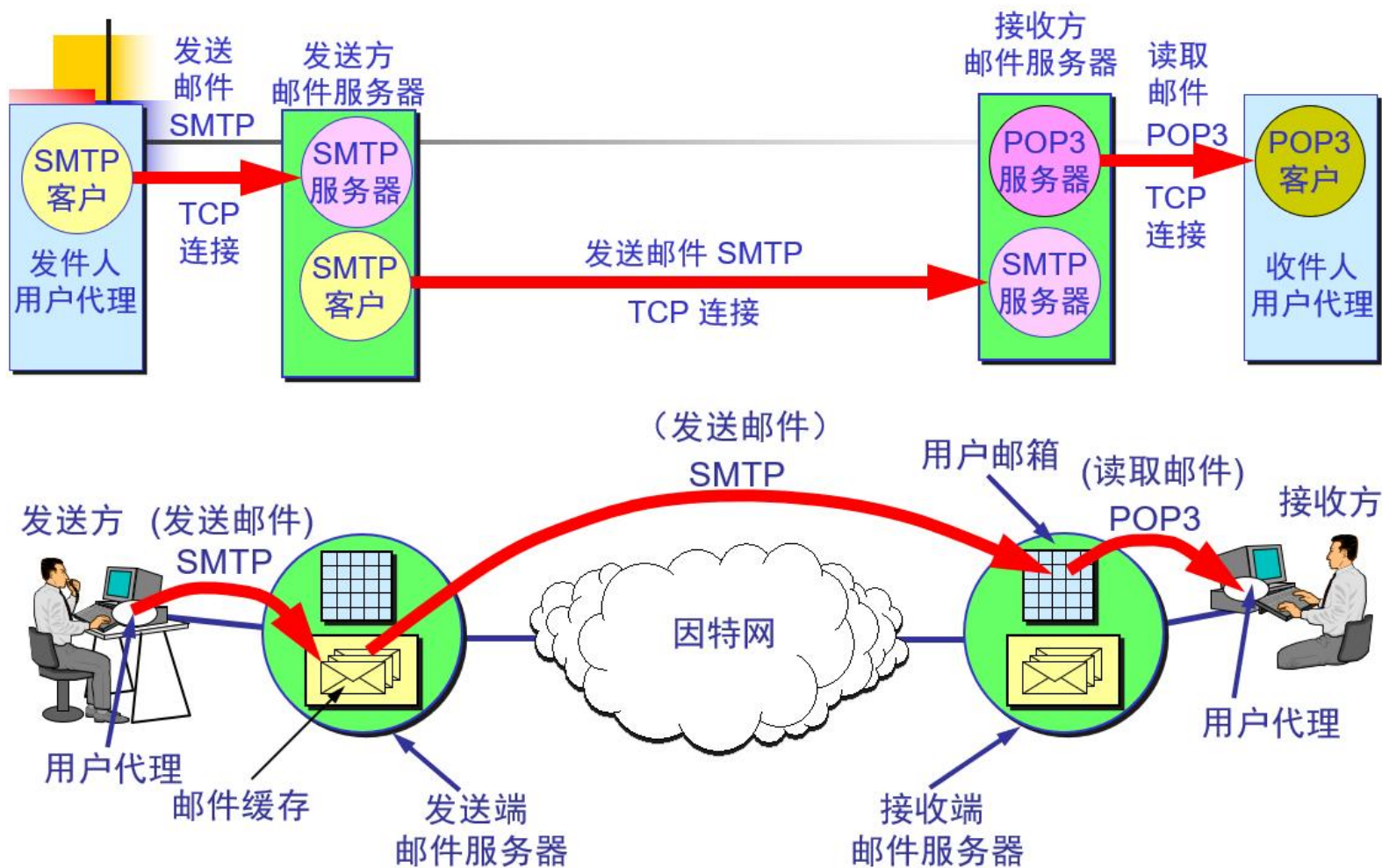
content-type: audio/basic
content-transfer-encoding: base64
--qwertyuiopasdfghjklzxcvbnm--

Example of a Multipart Message(2)

- Saving the above message in *.eml and opening with Foxmail



Mail transfer and delivery

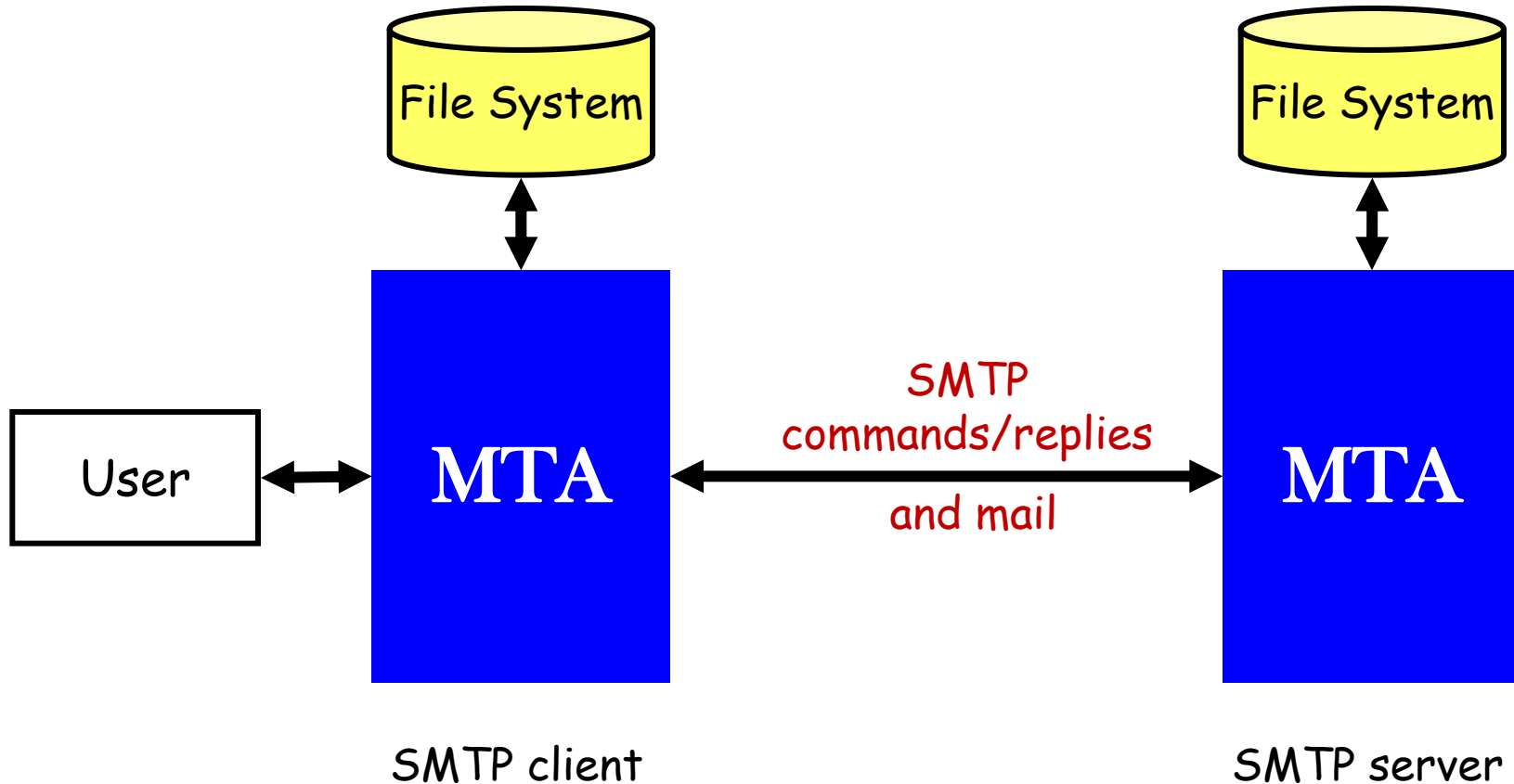


https://blog.csdn.net/qj_49780910

Mail transfer--SMTP

- Simple Mail Transfer Protocol: Used in transferring mail messages from UA to mail server, and between mail servers(MTA client and MTA server)
- SMTP is a simple **ASCII** protocol
- SMTP server listens to port **25**, SMTP client establishes a **TCP** connection to this port
- If the message cannot be delivered, an **error report** containing the first part of the undeliverable message is returned to the sender – eventually

SMTP Basic Model



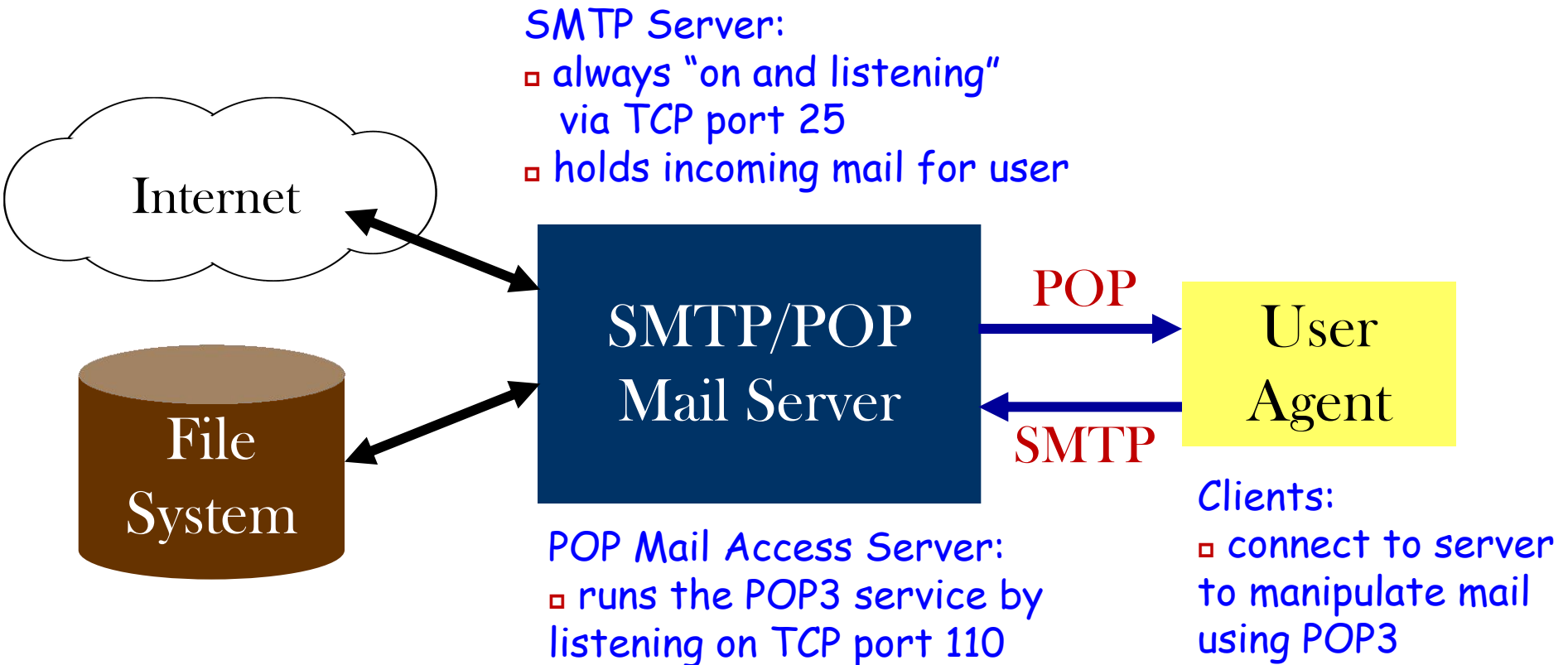
- MTA: Mail Transfer Agent, the mail server using SMTP

Example of SMTP messages

S: 220 ee.uwa.edu.au SMTP service ready C: <u>HELO</u> abcd.com S: 250 cs.washington.edu says hello to ee.uwa.edu.au C: <u>MAIL FROM:</u> <alice@cs.washington.edu> S: 250 sender ok C: <u>RCPT TO:</u> <bob@ee.uwa.edu.au> S: 250 recipient ok C: <u>DATA</u> S: 354 Send mail; end with "." on a line by itself C: From: alice@cs.washington.edu C: To: bob@ee.uwa.edu.au C: MIME-Version: 1.0 C: Message-Id: <0704760941.AA00747@ee.uwa.edu.au> C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm C: Subject: Earth orbits sun integral number of times C: C: This is the preamble. The user agent ignores it. Have a nice day. C: C: --qwertyuiopasdfghjklzxcvbnm C: Content-Type: text/html C: C: <p>Happy birthday to you C: Happy birthday to you C: Happy birthday dear <bold> Bob </bold> C: Happy birthday to you C: C: --qwertyuiopasdfghjklzxcvbnm C: Content-Type: message/external-body; C: access-type="anon-ftp"; C: site="bicycle.cs.washington.edu"; C: directory="pub"; C: name="birthday.snd" C: C: content-type: audio/basic C: content-transfer-encoding: base64 C: --qwertyuiopasdfghjklzxcvbnm C: S: 250 message accepted C: <u>QUIT</u> S: 221 ee.uwa.edu.au closing connection	Create connection Envelop Message (Header&Body) Close connection
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Mail delivery—POP: Basic Model

- Used to transfer mail from a mail server to a UA



POP: Features

- Essentially store and forward.
 - ◆ Mail is stored on the server until the client connects and then is downloaded to the client.
 - ◆ You MAY be able to leave a copy on the server
- Simple protocol and widely used
 - ◆ Many clients available such as foxmail, outlook
- protocol: POP3 (POP Version 3)

Using POP3 to Fetch 3 Messages

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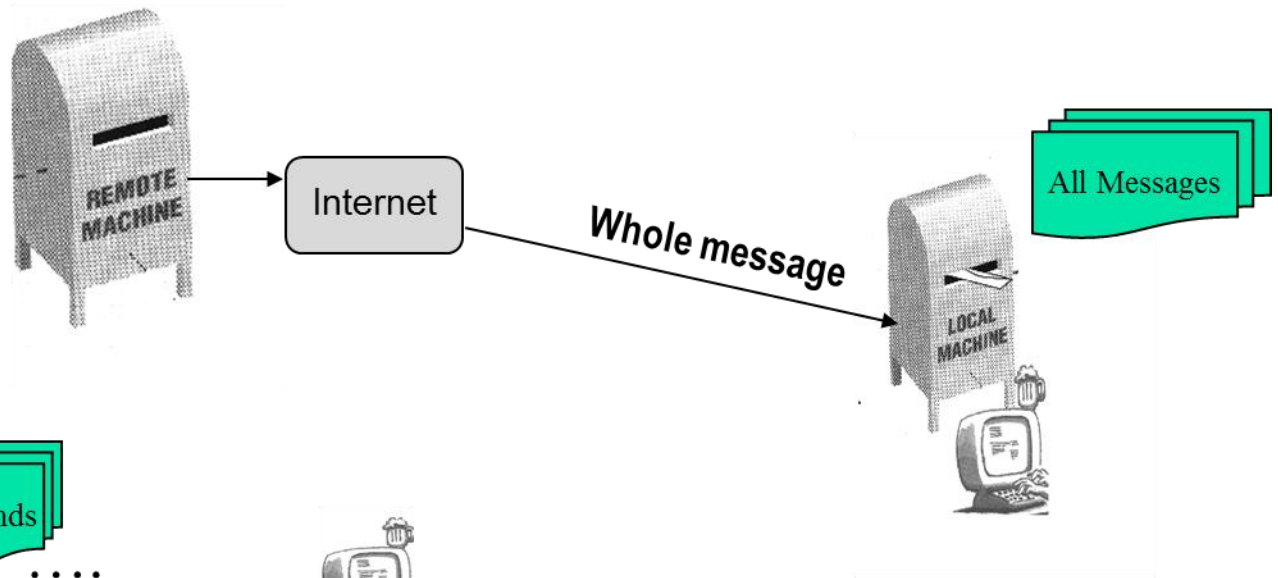
C: USER carolyn
S: +OK POP3 server ready
C: PASS vegetables
S: +OK
C: LIST
S: +OK login successful
C: LIST
S: 1 2505
S: 2 14302
S: 3 8122
S: .
C: RETR 1
S: (sends message 1)
C: DELE 1
C: RETR 2
S: (sends message 2)
C: DELE 2
C: RETR 3
S: (sends message 3)
C: DELE 3
C: QUIT
S: +OK POP3 server disconnecting
```

IMAP: Internet Message Access Protocol

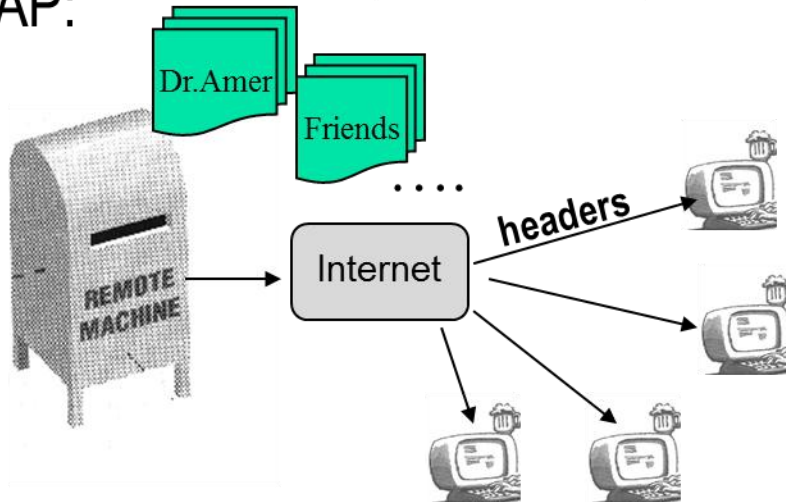
- POP3 is bad for mobile users or users that use multiple machines during the day
 - ◆ Mail is usually downloaded to the user agent computer, instead of remaining on the mail server.
 - ◆ Solution: IMAP (Internet Message Access Protocol)
- For mobile users or users that use multiple machines during the day
 - ◆ **IMAP** (Internet Message Access Protocol) is one of the main protocols that is used for **final delivery** of message.
 - ◆ RFC3501: IMAP4
 - ◆ The mail server runs an IMAP server that listens to port 143
 - ◆ The user agent runs an IMAP client.
 - ◆ IMAP is an improvement of POP3

POP3 vs IMAP(1)

POP3:



IMAP:

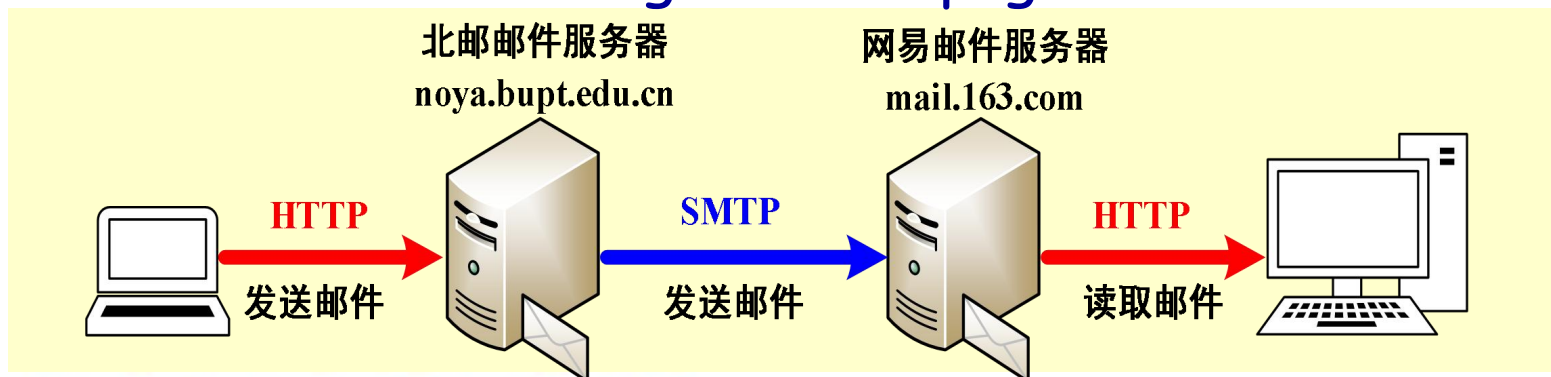


POP3 vs IMAP(2)

Feature	POP3	IMAP
Where is protocol defined?	RFC 1939	RFC 2060
Which TCP port is used?	110	143
Where is e-mail stored?	User's PC	Server
Where is e-mail read?	Off-line	On-line
Connect time required?	Little	Much
Use of server resources?	Minimal	Extensive
Multiple mailboxes?	No	Yes
Who backs up mailboxes?	User	ISP
Good for mobile users?	No	Yes
User control over downloading?	Little	Great
Partial message downloads?	No	Yes
Are disk quotas a problem?	No	Could be in time
Simple to implement?	Yes	No
Widespread support?	Yes	Growing

Webmail: HTTP

- Can deliver mail message in web page format



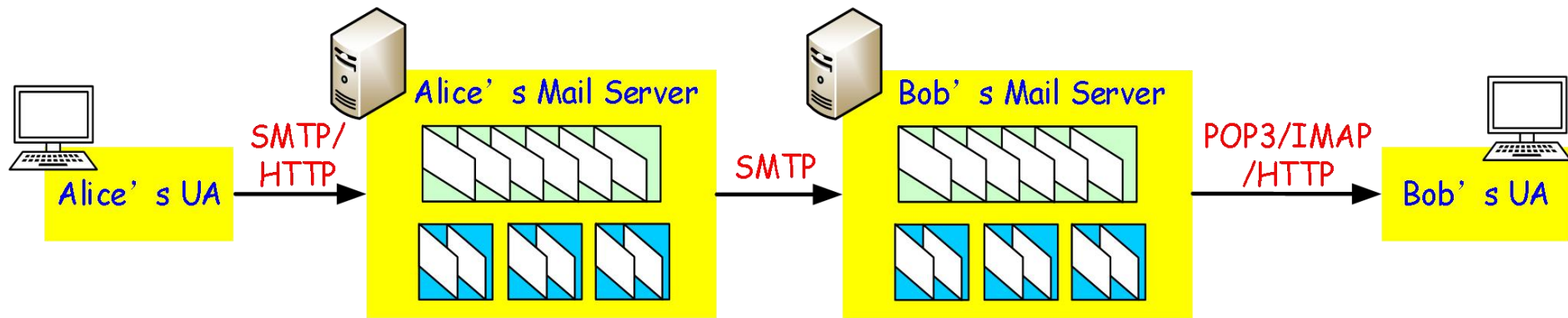
北京邮电大学邮件系统 <http://mail.bupt.edu.cn> [找回密码](#) [在线帮助](#)



The login interface for the BNU Webmail system. It includes the following elements:

- 登录入口 (Login Entry):** Indicated by an envelope icon.
- 用户名 (Username):** A text input field containing "chengli".
- @ bupt.edu.cn:** A fixed domain suffix.
- 密码 (Password):** A text input field with masked characters (dots).
- 记住用户名 (Remember Username):** A checked checkbox.
- SSL安全登录 (SSL Secure Login):** An unchecked checkbox.
- 登录 (Login):** A button to submit the login information.
- 注册 (Register):** A button to create a new account.

Protocols and communication entities



Review--Email

- 目标：用户之间的异步数据传输
- 邮件格式
 - ◆ 基本邮件格式(IMF)只支持ASCII（英文）
 - ◆ MIME：扩展邮件头，支持多种数据类型、多种语言、邮件内包含多部分(附件)
- 邮件系统的组成：UA，Mail server(MTA)，Mailbox
- 邮件传输协议(Client/Server)
 - ◆ 发送邮件协议：SMTP
 - ◆ 接收邮件协议：POP3, IMAP4(功能更强，更复杂)
- Webmail
 - ◆ 用户使用浏览器和HTTP协议与邮件服务器通信，收发邮件
 - ◆ 邮件服务器之间使用SMTP发送邮件
- 安全性(不讲)：PGP, SMIME

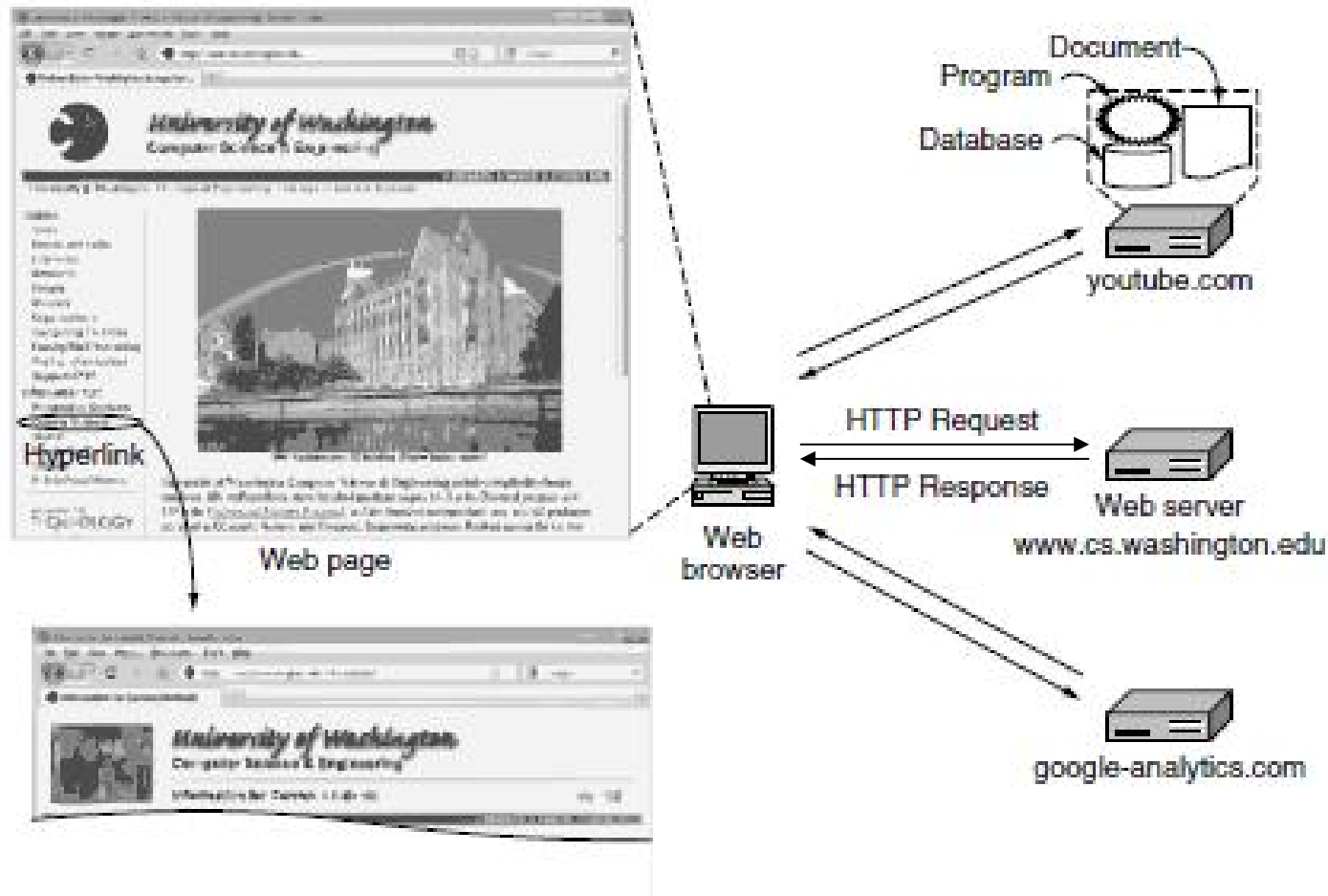
Outline

- Introduction to Application Layer
- Domain Name System (DNS)
- Email System and Protocols
- World Wide Web (WWW)
 - ◆ WWW Architecture
 - ◆ Hypertext Transfer Protocol (HTTP)

WWW: World Wide Web

- WWW(Web), one of mostly widely used Internet application, is an architectural framework for accessing linked content spread out over millions of machines all over the Internet.
 - ◆ Information is distributed in Internet, stored in web servers
 - ◆ accessed by hyperlinks(超级链接)
- A distributed hyper-media system, including multiple media information such as text, graph, image, audio, even video
- Client/Server mode, with browser program as client
- WWW documents are located with URL: Uniform Resource Locator
- WWW documents are written in HTML: HyperText Markup Language
- Client and server interact with HTTP: Hyper-Text Transfer Protocol
- Searching engines facilitate user to find information needed

Architectural Overview of Web



Web Client: Browser(浏览器)

- An application program used by the user to view the web page, as user's interface to Web
 - ◆ Fetching information (web page) from Web server
 - ◆ Displaying information for users
- After the user clicks on a hyperlink, Browser
 - ◆ Determines URL, name of the linked web page, eg. www.abc.com/example.html
 - ◆ Asks DNS for the IP address of www.abc.com
 - ◆ Establishes a TCP connection to replied IP address
 - ◆ Sends over an HTTP request asking for the page [/example.html](http://www.abc.com/example.html)
 - ◆ Fetches embedded images
 - ◆ Displays all the texts and images in [/example.html](http://www.abc.com/example.html)
 - ◆ Releases TCP connection

Web Server

- Stores set of Web documents
- Responds to request from browser by sending copy of document
- Steps of server's loop
 - ◆ Accept a **TCP connection** from a client (a browser).
 - ◆ Get the path to the page, which is the name of the file requested.
 - ◆ Get the file (from disk).
 - ◆ Send the contents of the file to the client.
 - ◆ Release the TCP connection

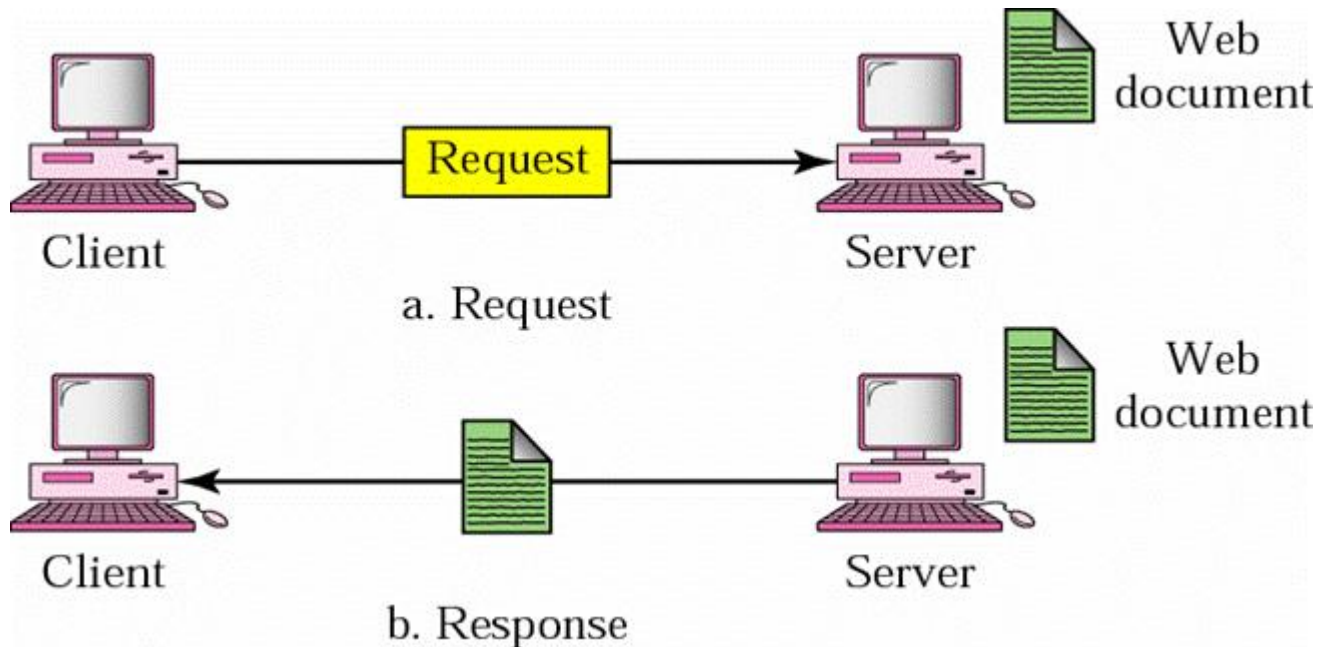
Web Documents

- Also called a **web page**(HTML)

- Type

 - ◆ **Static web page**

 - Each URL corresponded to a single file stored on some hard disk, **unchanging**
 - Edit in HTML format, .html, .htm



HTML – HyperText Markup Language

(a) The HTML for a sample Web page. (b) The formatted page.

```
<html>
<head><title> AMALGAMATED WIDGET, INC. </title> </head>
<body> <h1> Welcome to AWI's Home Page</h1>
 <br>
We are so happy that you have chosen to visit <b> Amalgamated Widget's </b>
home page. We hope <i> you </i> will find all the information you need here.
<p>Below we have links to information about our many fine products.
You can order electronically (by WWW), by telephone, or by fax. </p>
<hr>
<h2> Product information </h2>
<ul>
  <li> <a href="http://widget.com/products/big"> Big widgets</a>
  <li> <a href="http://widget.com/products/little"> Little widgets </a>
</ul>
<h2> Telephone numbers</h2>
<ul>
  <li> By telephone: 1-800-WIDGETS
  <li> By fax: 1-415-765-4321
</ul>
</body>
</html>
```

(a)

Welcome to AWI's Home Page



We are so happy that you have chosen to visit **Amalgamated Widget's** home page. We hope *you* will find all the information you need here.

Below we have links to information about our many fine products. You can order electronically (by WWW), by telephone, or by FAX.

Product Information

- [Big widgets](#)
- [Little widgets](#)

Telephone numbers

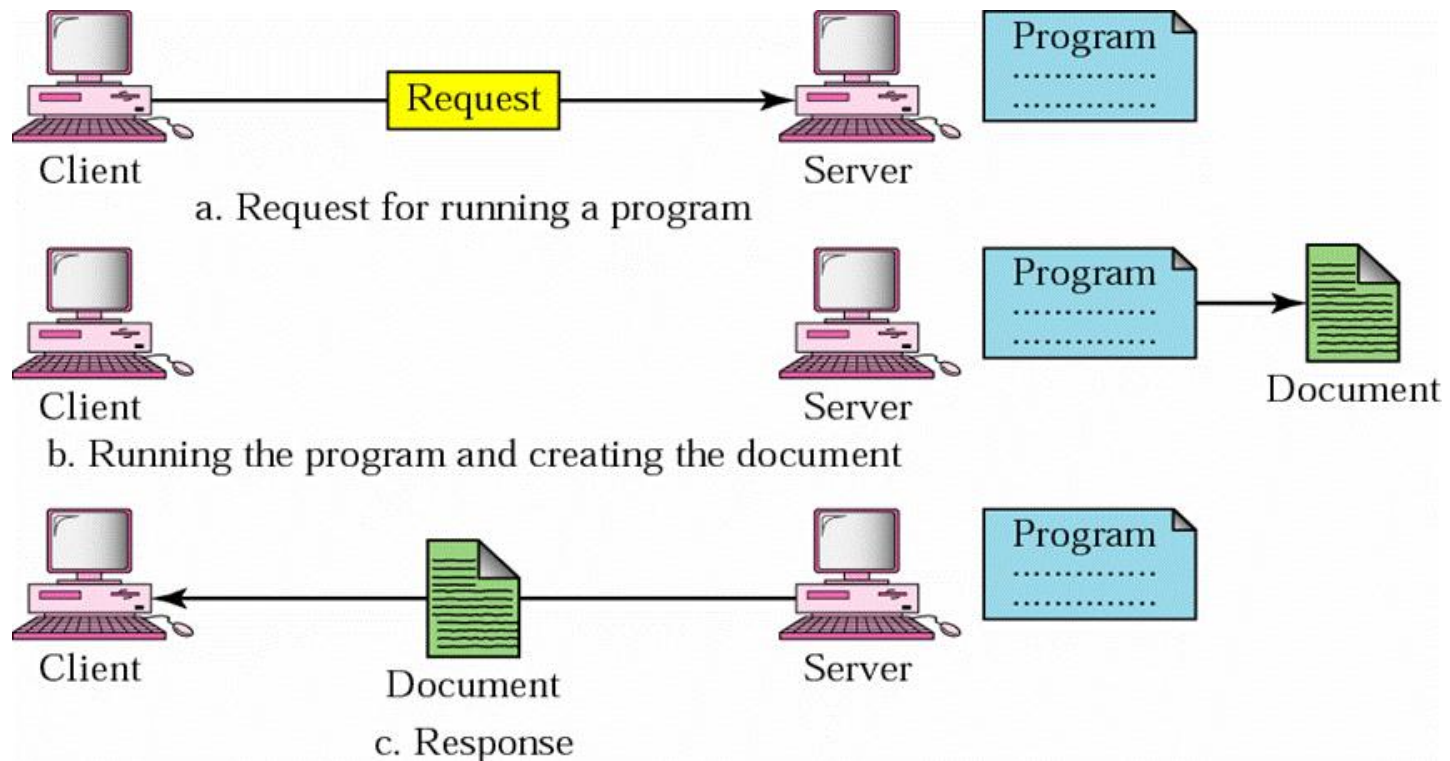
- 1-800-WIDGETS
- 1-415-765-4321

(b)

Web Documents

◆ Dynamic web page

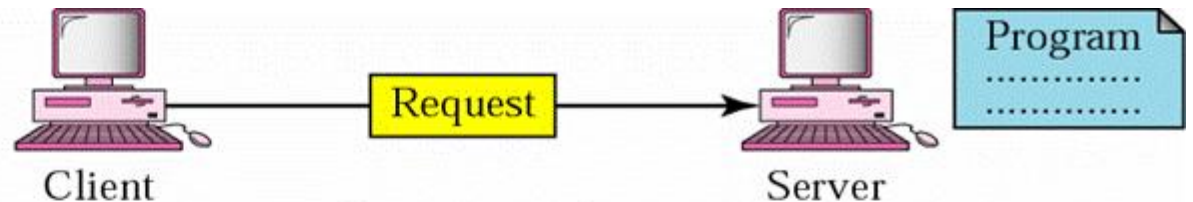
- Can **change** whenever a server receives a request
- **Generated on server side**
- Output from a program on server, usually a script, ASP, JSP, VB Script, PHP, CGI or other programs
- .asp, .shtm, .php, .cgi etc.



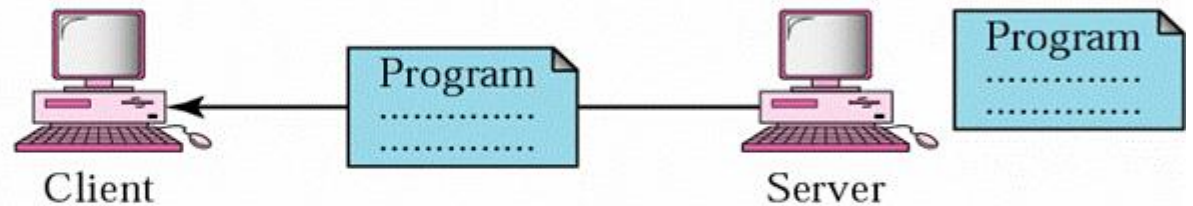
Web Documents

◆ Active web page

- Can **change** after the document has been **loaded** into a browser
- **Generated on client side**
- Consists of a computer program, known as Java, running in browser



a. Request for a copy of a program



b. Sending a copy of the program



c. Running the program and creating the document

Links to Other Pages

- Embedded in HTML document
- Browser
 - ◆ Hides text of link from user
 - ◆ Associates link with item on page
- Called Uniform Resource Locator (**URL**)统一资源定位符
 - ◆ Serving as worldwide name of a web page
 - ◆ Consists of three parts: the **protocol**(*http/https*); the **DNS name of the host**(*www.bupt.edu.cn*); the **path name**(*index.html*)

https :// www.bupt.edu.cn / index.html

protocol :// **domain_name** : **port** / **path_name**

The diagram illustrates the components of the URL `https://www.bupt.edu.cn/index.html`. Brackets are used to group parts of the URL and link them to descriptive labels below. The `https` part is bracketed and labeled 'name of access protocol to use'. The `://` part is bracketed and labeled 'domain name of server computer'. The `www.bupt.edu.cn` part is bracketed and labeled 'protocol port number'. The `/` part is bracketed and labeled 'path name of item'. The `index.html` part is bracketed and labeled 'path name of item'.

name of access
protocol to use

domain name of
server computer

protocol port
number

path name
of item

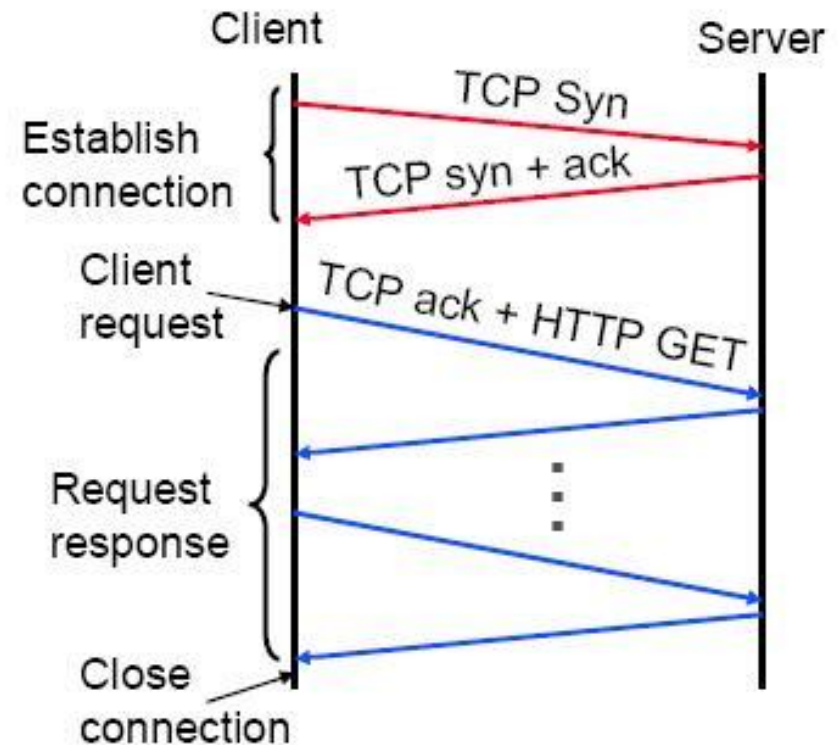
Some Common URLs

- Web browsers are often called Universal Clients because most can talk other protocols besides HTTP

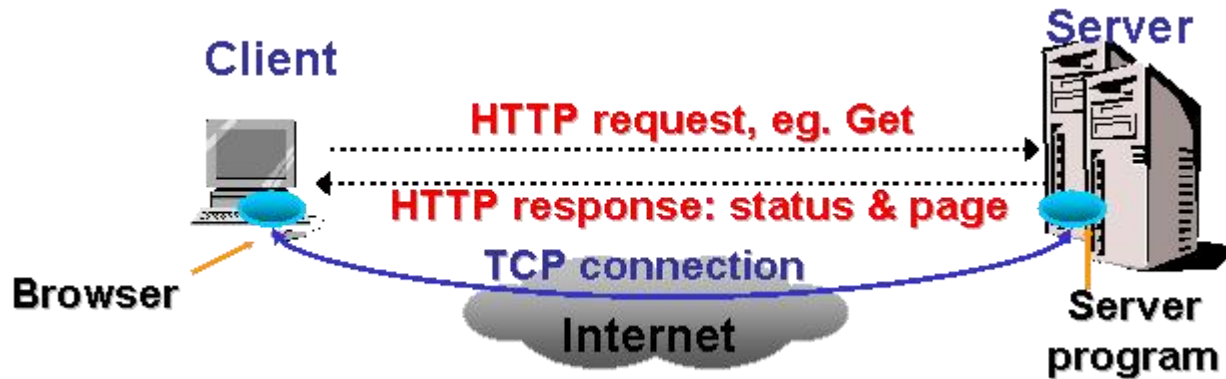
Name	Used for	Example
http	Hypertext (HTML)	http://www.ee.uwa.edu/~rob/
https	Hypertext with security	https://www.bank.com/accounts/
ftp	FTP	ftp://ftp.cs.vu.nl/pub/minix/README
file	Local file	file:///usr/suzanne/prog.c
mailto	Sending email	mailto:JohnUser@acm.org
rtsp	Streaming media	rtsp://youtube.com/montypython.mpg
sip	Multimedia calls	sip:eve@adversary.com
about	Browser information	about:plugins

HTTP : Hyper-Text Transfer Protocol

- Client-server architecture
- Synchronous request/reply protocol
 - ◆ Running over TCP
 - ◆ Port 80(http),443(https)
- Stateless
 - ◆ Each command is executed independently, without any knowledge of the previous operations

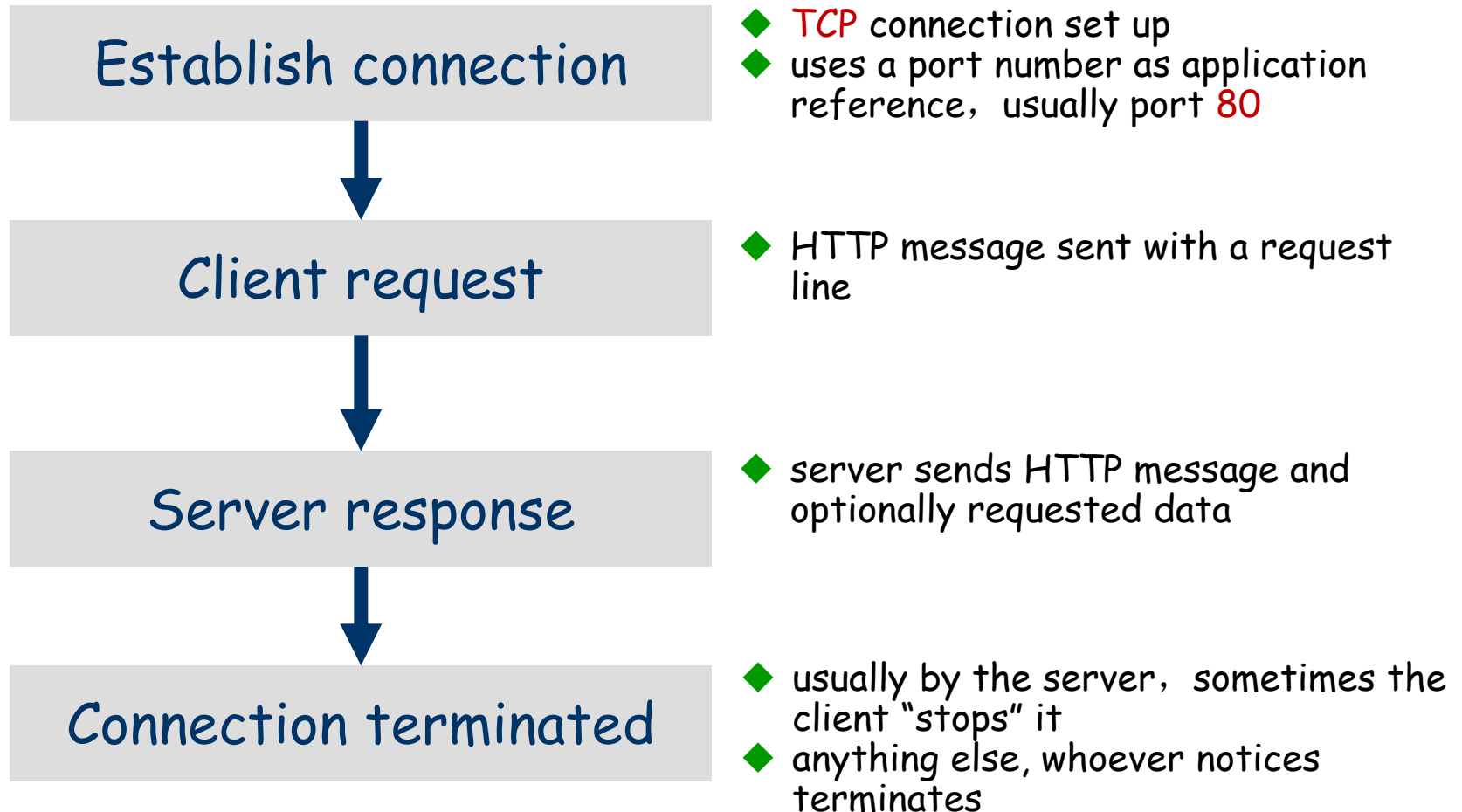


Operations of HTTP

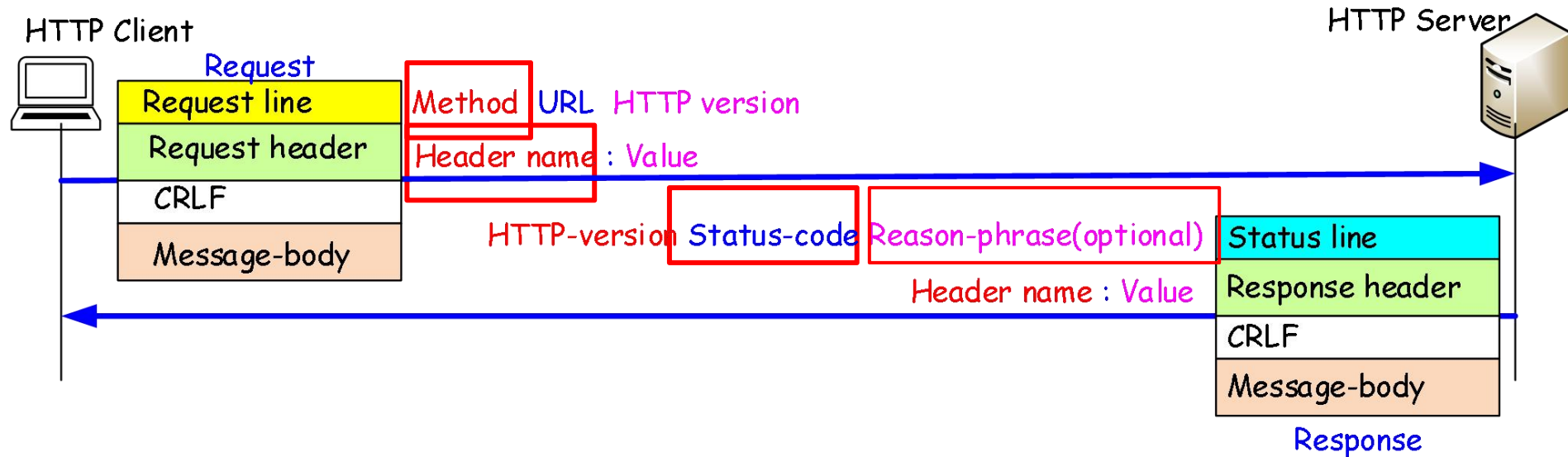


- (1) Browser **analyses URL** of the hyperlink, i.e. `http://www.abc.com/example.html`
- (2) Browser **uses DNS to get IP address** of the server (`www.abc.com`)
- (3) Browser sets up a **TCP connection** with server
- (4) Browser sends a **HTTP request: GET** /example.html
HTTP/1.0
- (5) Server sends **requested page** to browser
- (6) **TCP connection is released**
- (7) Browser **displays** example.html

HTTP Transaction



HTTP Transaction



HTTP Request Methods

Method	Description
GET	Request to read a Web page
HEAD	Request to read a Web page's header
PUT	Request to store a Web page
POST	Append to a named resource (e.g., a Web page)
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Reserved for future use
OPTIONS	Query certain options

HTTP Message Headers

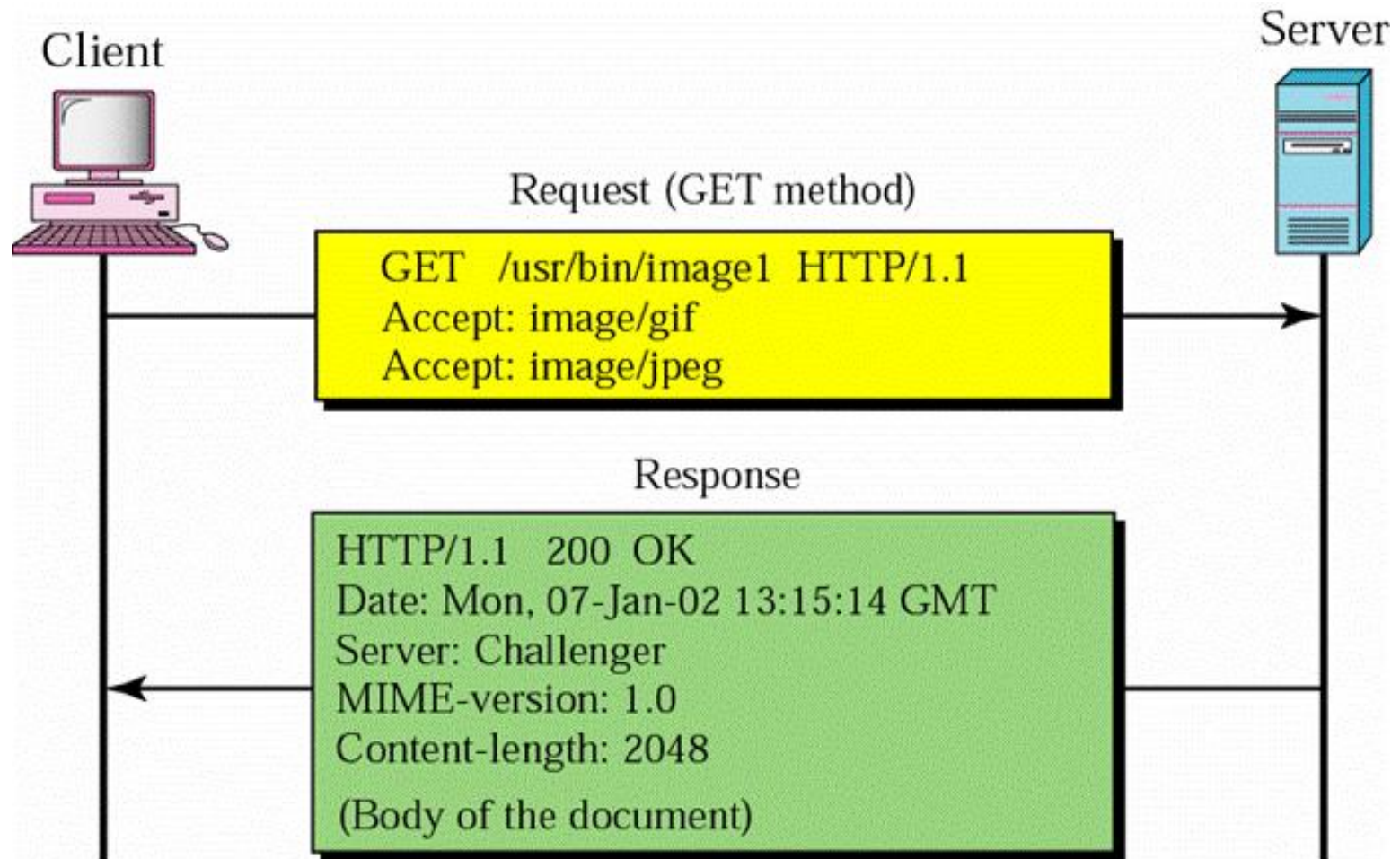
Header	Type	Contents
User-Agent	Request	Information about the browser and its platform
Accept	Request	The type of pages the client can handle
Accept-Charset	Request	The character sets that are acceptable to the client
Accept-Encoding	Request	The page encodings the client can handle
Accept-Language	Request	The natural languages the client can handle
Host	Request	The server's DNS name
Authorization	Request	A list of the client's credentials
Cookie	Request	Sends a previously set cookie back to the server
Date	Both	Date and time the message was sent
Upgrade	Both	The protocol the sender wants to switch to
Server	Response	Information about the server
Content-Encoding	Response	How the content is encoded (e.g., gzip)
Content-Language	Response	The natural language used in the page
Content-Length	Response	The page's length in bytes
Content-Type	Response	The page's MIME type
Last-Modified	Response	Time and date the page was last changed
Location	Response	A command to the client to send its request elsewhere
Accept-Ranges	Response	The server will accept byte range requests
Set-Cookie	Response	The server wants the client to save a cookie

HTTP Status Codes and reason-phrase

Code	Meaning	Examples
1xx	Information	100 = server agrees to handle client's request
2xx	Success	200 = request succeeded; 204 = no content present
3xx	Redirection	301 = page moved; 304 = cached page still valid
4xx	Client error	403 = forbidden page; 404 = page not found
5xx	Server error	500 = internal server error; 503 = try again later

Status code	Reason phrase
100	Continue
200	OK
204	No content
301	Moved permanently
304	Not modified
403	Forbidden
404	Not found
500	Internal server error
503	Service unavailble

Example of HTTP Transaction



[Forouzan]

HTTP/1.1

Performance Enhancements

- HTTP/1.0 is a “stop and wait” protocol
 - ◆ Separate TCP connection for each file
 - Connect setup and tear down is incurred for each file
 - Inefficient use of packets
 - Server must maintain many connections
- HTTP/1.1 specification focus on performance enhancements
 - ◆ Persistent connections
 - ◆ Pipelining
 - ◆ Enhanced caching options
 - ◆ Support for compression

HTTP/1.1

Persistent Connections and Pipelining

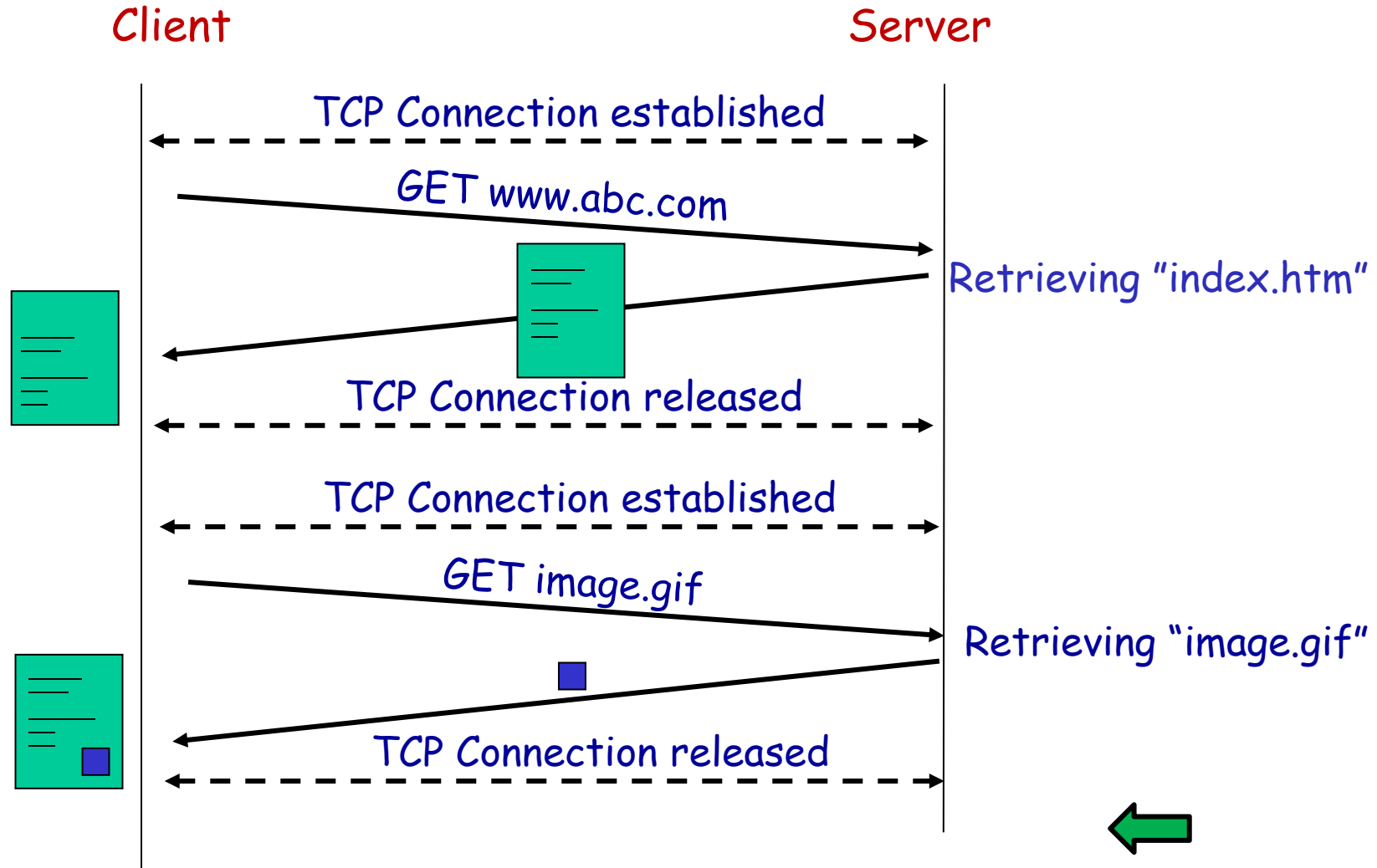
■ Persistent connections(持续连接)

- ◆ Use the same TCP connection(s) for transfer of multiple files
- ◆ Reduces packet traffic significantly

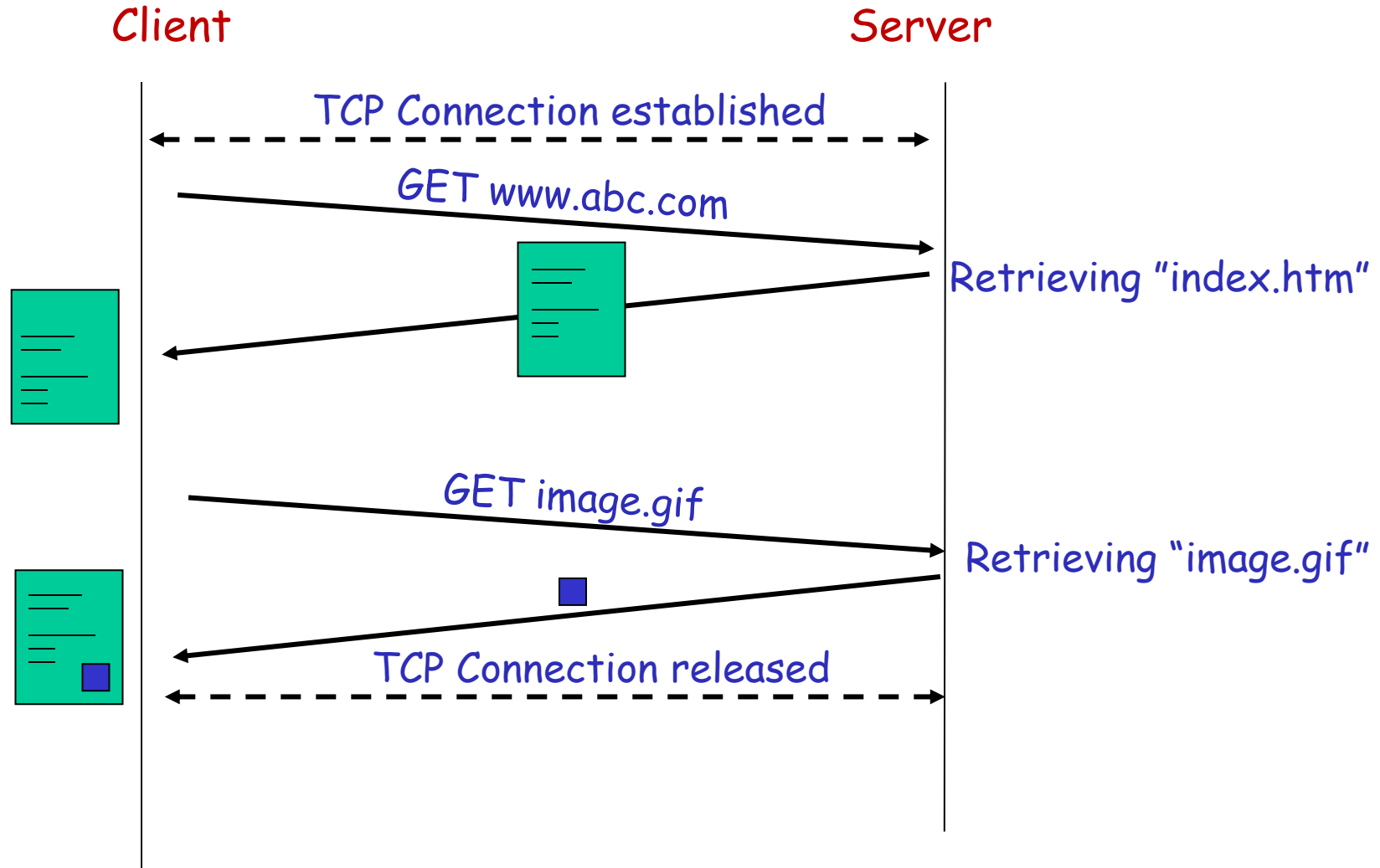
■ Pipelining(流水线)

- ◆ **Multiple HTTP requests** can be written out to a socket together without waiting for the corresponding responses.
- ◆ Pack several HTTP requests into one TCP/IP packet

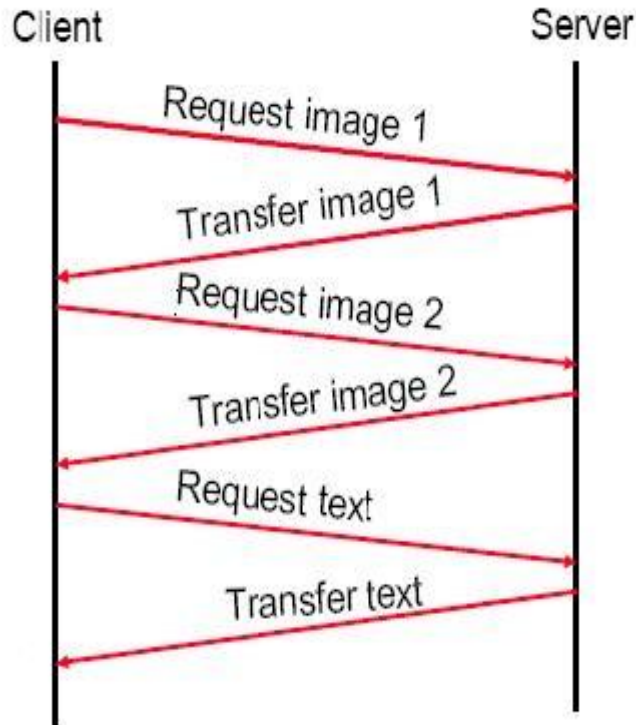
Example of Non-persistent connections



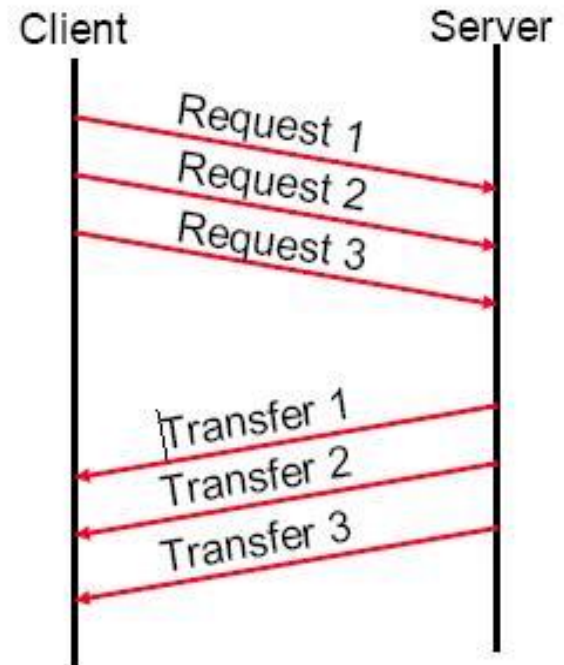
Example of Persistent connections



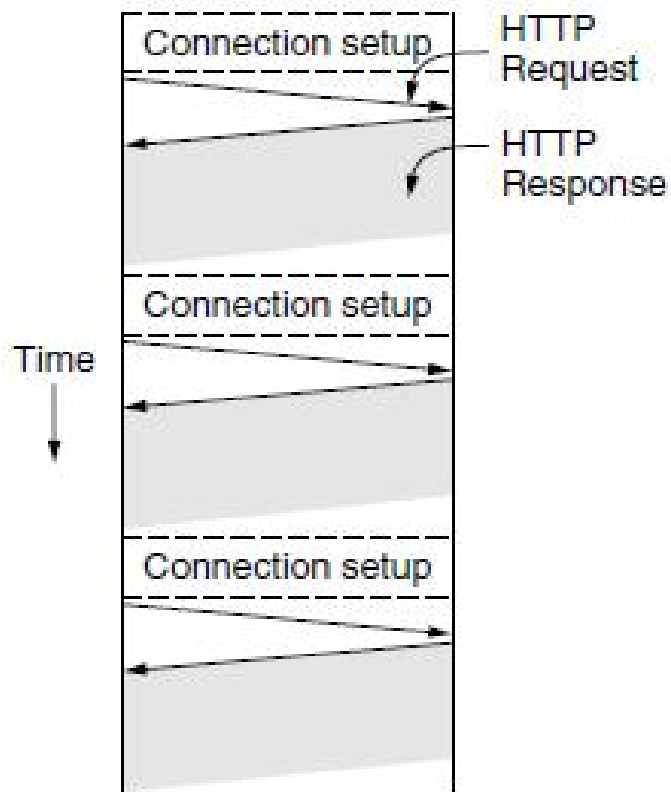
Example of Pipelining(流水线)



Non-pipelining

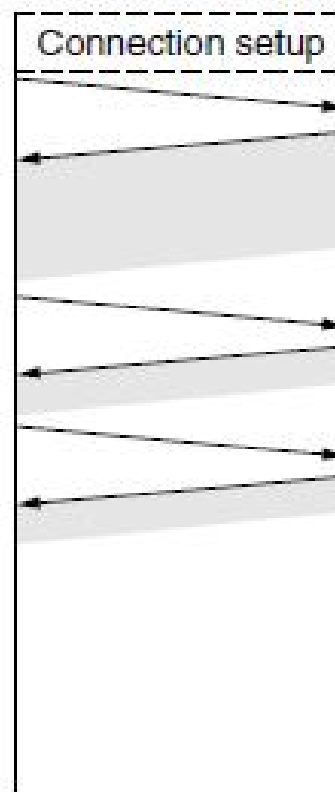


Pipelining



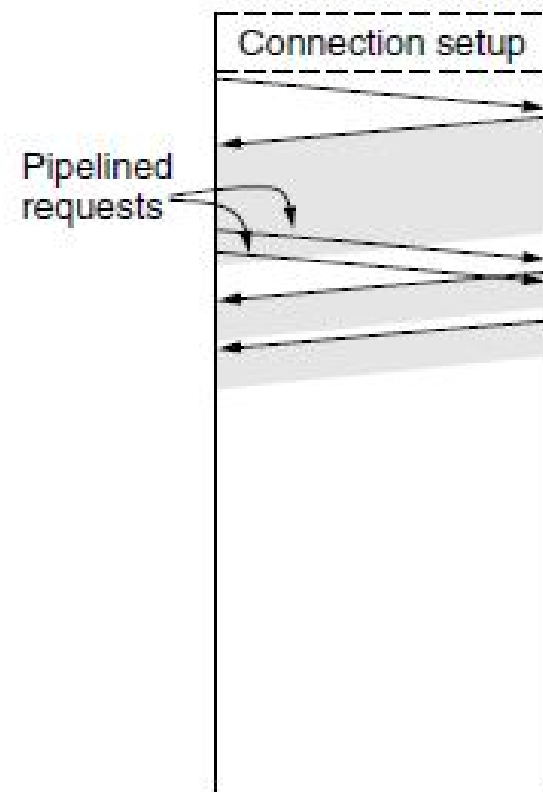
(a)

multiple connections
and sequential requests



(b)

A persistent
connection and
sequential requests



(c)

A persistent
connection and
pipelined requests

Review--WWW

- 目标：将分布在互联网不同设备上的资源组织起来。
- WWW体系结构
 - ◆ 超链接：超文本、超媒体
 - ◆ Web客户(浏览器)和Web服务器
 - ◆ URL: `protocol://domain_name:port/item_name`
- Web页面：静态页面(HTML)，动态页面，活动页面
- 超文本传输协议(HTTP)
 - ◆ Client/Server, 同步数据传输
 - ◆ over TCP, port 80
 - ◆ HTTP操作：DNS, TCP连接, HTTP请求报文和响应报文
 - ◆ HTTP1.0&1.1: 持续链接，流水线请求
- HTTP安全(不讲): HTTPS (HTTP over SSL/TLS)

Key Points(1)

- Idea of client/server model
- DNS
 - ◆ Domain name, hierarchical architecture
 - ◆ Jobs of resolver and DNS servers
 - ◆ Iterative resolution vs. Recursive resolution
- WWW
 - ◆ Concepts of WWW, HTTP, HTML, hyperlink, URL, browser
 - ◆ Operations of HTTP
 - ◆ HTTP 1.0 vs. HTTP 1.1

Key Points(2)

■ Email

- ◆ Email address
- ◆ SMTP: sending Email from user machine to receiver's mailbox
- ◆ POP3/IMAP: downloading Email from mailbox to user machine
- ◆ Webmail: using HTTP

复习1—协议

- 应用层：HTTP, DNS, SMTP, POP3, IMAP4, MIME, DHCP, RIP, BGP
- 传输层：TCP, UDP
- 网络层：IP, ICMP, OSPF, ARP/RARP
- 数据链路层：PPP, HDLC, IEEE802.3, IEEE802.11

大家需要知道不同的应用层协议的底层是UDP还是TCP

复习2—设备

- 上层：主机、网关
- 网络层：路由器，**NAT**盒子
- 数据链路层：交换机(直通式，存储-转发式)、网桥
- 物理层：中继器、集线器、**ADSL**、**Modem**、**DSLAM**、编解码器、复用器

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